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Design of an Automated Electronic Greenhouse

Mechatronic Project 478
Final Report

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2025/07/07

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Executive summary

Title of Project
The Design of an Automated, Solar-Powered Greenhouse for Optimal Plant Growth
Objectives
Design and build a small-scale greenhouse capable of automating all the necessary elements to sustain amicable living conditions for various plants.
What is current practice and what are its limitations?
Current domestically applicable greenhouses rely on allot of active monitoring and care, this takes up allot of time and can still not achieve ideal living conditions for some plants.
What is new in this project?
This project introduces the monitoring of various parameters within the greenhouse such as temperature, humidity, soil moisture air composition and airflow and allows the effective controlling of these parameters to a desired range set by the user. This project also introduces solar power to offset the power requirements reinforce the sustainability.
If the project is successful, how will it make a difference?
It will introduce the ease of automation to domestic greenhouses, demystifying the struggles of home farming while encouraging self-sufficiency. By promoting food independence, it can reduce reliance on the national food supply chain, empowering individuals and communities to grow their own fresh produce sustainably.
What are the risks to the project being a success? Why is it expected to be successful?
The system may not be efficient enough at controlling the desired parameters. There may be delays due to time management issues. However, through thorough research, modular design and allowing buffer periods for unexpected delays, the project shouldn't be able to fail.
What contributions have/will other students made/make?
No other students have/ will contribute to this design.
Which aspects of the project will carry on after completion and why?
Further research on the economic feasibility of the project may be conducted.
What arrangements have been/will be made to expedite continuation?
Thorough documentation of the design process will allow for continuation or improvements to be made. Modular design should allow for ease of maintenance as well as allow for the upgrading of any components if necessary.

Acknowledgements

I would like to begin by sincerely thanking my supervisor for proposing a project that I feel deeply passionate about, and for providing the guidance and support that allowed it to grow into what it has become. I also wish to acknowledge all the staff in the M&M Engineering Department at Stellenbosch University who assisted in many aspects of the project. In no particular order: Mr Cobus and Ferdi Zietsman, Mr Llewellyn Kordom, Mr Nathi HLwempu, Mr Kevin Neaves, Charl Adonis, Mrs. Paulse Simorne and Mr Cobus Samuels. This project could not have taken the shape it did without their dedication, expertise, and passion for engineering and their respective crafts.

ECSA Outcomes Self-Assessment

ECSA Outcome	Addressed in Sections:
GA 1. Problem solving: Demonstrate competence to identify, assess, formulate and solve convergent and divergent engineering problems creatively and innovatively.	• Concept Design Section 6
GA 2. Application of scientific and engineering knowledge: Demonstrate competence to apply knowledge of mathematics, basic science and engineering sciences from first principles to solve engineering problems.	• Calculations Appendix D
GA 3. Engineering Design: Demonstrate competence to perform creative, procedural and non-procedural design and synthesis of components, systems, engineering works, products or processes.	• Concept Design Section 6 • Detailed Design Section 7
GA 5. Engineering methods, skills and tools, including Information Technology: Demonstrate competence to use appropriate engineering methods, skills and tools, including those based on information technology.	• Design Evaluation Section 9
GA 6. Professional and technical communication: Demonstrate competence to communicate effectively, both orally and in writing, with engineering audiences and the community at large.	• Project Proposal • Progress Report / Presentation • Safety Report • First Draft / Final Report • Final Presentation
GA 8. Individual, Team and Multidisciplinary Working: Demonstrate competence to work effectively as an individual, in teams and in multi-disciplinary environments.	• Safety Report Section B • Provided Equipment Section 7.8.4
GA 9. Independent Learning Ability: Demonstrate competence to engage in independent learning through well-developed learning skills.	• Literature Review and Market Research Section 5 • Detailed Design Section 7

Table 2: ECSA Graduate Attributes Mapping

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Variables

I	Moment of Inertia	[kgm ²]
E	Youngs Modulus	[Pa]
ν	Poisons Ratio	[]
P	Dynamic Pressure	[Pa]
D	Flexural Rigidity	[Nm]
ρ	Density	[kg/m ³]

Vectors and Tensors

$\vec{v}$	Velocity
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Subscripts

h	Height
t	Thickness
l	Length
b	Breadth

Abbreviations

PAR	Photosynthetic Active Radiation
PPFD	Photosynthetic Photon Flux Density
DLI	Daily Light Integral
VWC	Volumetric Water Content
HVAC	Heating Ventilation Air Conditioning
CO ₂	Carbon Dioxide
ppm	parts per million
IoT	Internet of Things
IDE	Integrated Development Environment
TCP	Transmission Control Protocol

IP	Internet Protocol
PV	Photo Voltaic
DC	Direct Current
AC	Alternating Current
MPPT	Maximum Power Point Tracking
RH	Relative Humidity
GPIO	General Purpose Input Output
CFM	Cubic Feet per Minute
DoD	Depth of Discharge

4 Introduction

4.1 Background

As food insecurity increases and available farmland decreases due to deforestation and urbanization (Olagunju, 2015), small-scale greenhouses offer a practical way to grow food in limited spaces. Greenhouses help protect crops from harsh weather and allow for better control of growing conditions, which can improve yield, product quality and lengthen market availability (Vox *et al.*, 2010). By using transparent materials to trap heat and light, they create a more stable environment for plants. However, maintaining this environment can be labour intensive and often requires regular monitoring to manage temperature, humidity, and watering.

4.2 Objectives

- Design a small-scale greenhouse that can monitor and adjust key environmental factors such as temperature, soil moisture, and humidity.
- Integrate wireless data logging with a dashboard to enable users to access the data from anywhere in real time.
- Allow users to input and control target conditions based on the specific requirements of different plant types.
- Integrate a solar power system to enable off-grid operation.
- Test and evaluate the performance of the greenhouse system to assess its effectiveness and reliability in maintaining suitable growing conditions.

4.3 Motivation

Greenhouses provide ideal growing conditions but require constant monitoring and adjustments, making them labour-intensive and inaccessible for many. By reducing manual effort and enabling customization for different plants, this project aims to empower users to grow food efficiently and sustainably with minimal effort. Additionally, the off-grid solar design promotes energy independence and environmental sustainability, aligning with global efforts to combat food insecurity and climate change. This project bridges technology and agriculture, making greenhouse farming accessible, efficient, and eco-friendly.

4.4 Methodology

This project adopts the engineering design methodology outlined by Dieter and Schmidt, providing a structured framework to guide the development of the electronic greenhouse. (Dieter and Schmidt) This methodology encompasses several key phases:

1. **Problem Definition and Need Identification:** This initial phase emphasizes understanding the problem's scope and identifying the specific needs the design aims to fulfill.
2. **Concept Generation:** Developing multiple design concepts that address the identified needs and constraints. Multiple potential solutions are brainstormed without immediate judgment, encouraging a broad exploration of ideas.
3. **Concept Selection:** Here, the various concepts are evaluated against predefined criteria to select the most promising solution.
4. **Detailed Design:** Precise specifications for all components are established, including materials, dimensions, and tolerances, preparing the design for manufacturing.
5. **Design Evaluation:** Testing the designed system to determine its functionality and performance.

By following this systematic approach, the project ensures a thorough exploration of design alternatives and a robust development process for the electronic greenhouse.

4.5 Use of AI Tools

AI tools were used only in the writing of this report to enhance language clarity and grammar, ensuring technical information is conveyed effectively.

5 Literature review and market research

5.1 Environmental Requirements for Optimal Plant Growth

Different plant species require a careful balance of light, temperature, humidity, and water supply to achieve optimal growth. The following sections outline the key environmental parameters that must be monitored and controlled to create ideal growing conditions within the greenhouse.

5.1.1 Temperature

Plants regulate transpiration and cooling through specialized organs known as stomata located on the surface of leaves. These pores can open or close to control the release of water vapor, helping the plant manage water loss and internal temperature (CANNA South Africa, n.d.). As ambient temperatures rise, open stomata allow for increased evaporation, aiding in the cooling process. However, excessive heat can lead to excessive water loss and stress. While optimal temperature ranges vary depending on the plant species and its growth stage, greenhouse temperatures are generally best maintained between 15 °C and 30 °C to support healthy plant development (Koşan *et al.*, 2020).

5.1.2 Light

Light indirectly influences water loss in plants by triggering the opening of stomata, which enables gas exchange and CO₂ uptake for photosynthesis (Kochetova *et al.*, 2022). However, excess light can cause overheating and increased transpiration, so it must be carefully managed along with temperature. Some plants are sensitive to photoperiodism, meaning their flowering depends on day length. In greenhouses, adjusting light exposure can control flowering times, allowing for year-round yield (Torres *et al.*).

To evaluate light conditions within greenhouses, specific terminology and measurement units are used to describe both the quantity and quality of light that plants receive. The spectrum of light between 400 and 700 nanometres (nm) is recognized as **photosynthetically active radiation (PAR)**, the range most efficiently used by plants for photosynthesis (Sena *et al.*, 2024). Monitoring this specific light range is essential for optimizing plant productivity.

A key metric within the PAR range is the **photosynthetic photon flux density (PPFD)**. PPFD refers to the number of photons, within the PAR spectrum, that strike a square meter of plant surface each second. It is expressed in micromoles per square meter per second ($\mu\text{mol}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$). PPFD measurements help determine if the lighting levels in the greenhouse are sufficient to support healthy plant growth. Insufficient PPFD may lead to slow growth or poor productivity due to inadequate light. Excessively high PPFD can result in photoinhibition or physical damage to plant tissues (Hortitech Help Center, n.d.).

To relate PPFD to the more commonly used unit of *lux*, an approximate conversion can be applied. The exact factor depends on the spectrum of the light source, but for white LEDs, a typical approximation is:

$$\text{PPFD } (\mu\text{mol}\cdot\text{m}^{-2}\cdot\text{s}^{-1}) \approx \frac{\text{Illuminance (lux)}}{54} \quad (5.1)$$

where $1 \mu\text{mol}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$ corresponds to roughly 54 lux. This relationship allows lux sensor readings to be interpreted in terms of usable light for photosynthesis. (Ahn *et al.*, 2017)

Monitoring and balancing light intensity through artificial lighting systems and proper greenhouse design is therefore essential to maintain optimal photosynthetic activity without causing plant stress.

5.1.3 Humidity

Humidity plays a critical role in plant health by influencing the rate of transpiration, the process through which plants absorb water and nutrients from the soil and release water vapor through stomata. In high humidity environments, the air becomes too saturated with moisture, which suppresses transpiration. This reduction in water and nutrient uptake can hinder plant growth and development.

Conversely, in low humidity conditions, transpiration rates can become excessively high due to the dry air. In response, plants often close their stomata to conserve water, but this also limits gas exchange and light absorption, leading to water stress and reduced growth (Nederhoff, n.d.).

To support optimal development, research indicates that the ideal relative humidity (RH) range for most greenhouse crops lies between 65% and 85%. Maintaining this balance helps prevent both under- and over-transpiration, ensuring steady nutrient uptake and healthy physiological function (Drygair Greenhouse Dehumidifiers, n.d.).

5.1.4 Carbon Dioxide

For most greenhouse crops, net photosynthesis increases significantly as CO_2 concentrations rise from ambient levels (approximately 340 ppm) up to around 1,000 ppm, potentially boosting photosynthesis by up to 50%. However, the benefit of CO_2 supplementation varies by crop and may not be economically justifiable under low light conditions or for certain species like tulips and Easter lilies, which show little response (UC Agriculture and Natural Resources, n.d.).

CO_2 enters plants through stomata via diffusion, and its uptake is influenced by external CO_2 concentration as well as factors such as light, temperature, humidity, and water availability. While plants generally grow well at ambient CO_2 levels (about 340 ppm), concentrations can drop to as low as 200 ppm in sealed greenhouses during the day due to plant uptake. This drop can significantly reduce photosynthesis, sometimes with an effect comparable to the enhancement achieved by raising CO_2 levels to 1,300 ppm.

As seen in the figure 5.4, at low CO_2 concentrations (below approximately 300 ppm), the rate of photosynthesis increases sharply as CO_2 concentration rises. However, beyond approximately 400–500 ppm (close to ambient fresh air levels), further increases in CO_2 concentration have little to no additional effect on the photosynthesis rate, as the curve begins to plateau. Therefore, monitoring and maintaining ambient CO_2 concentrations

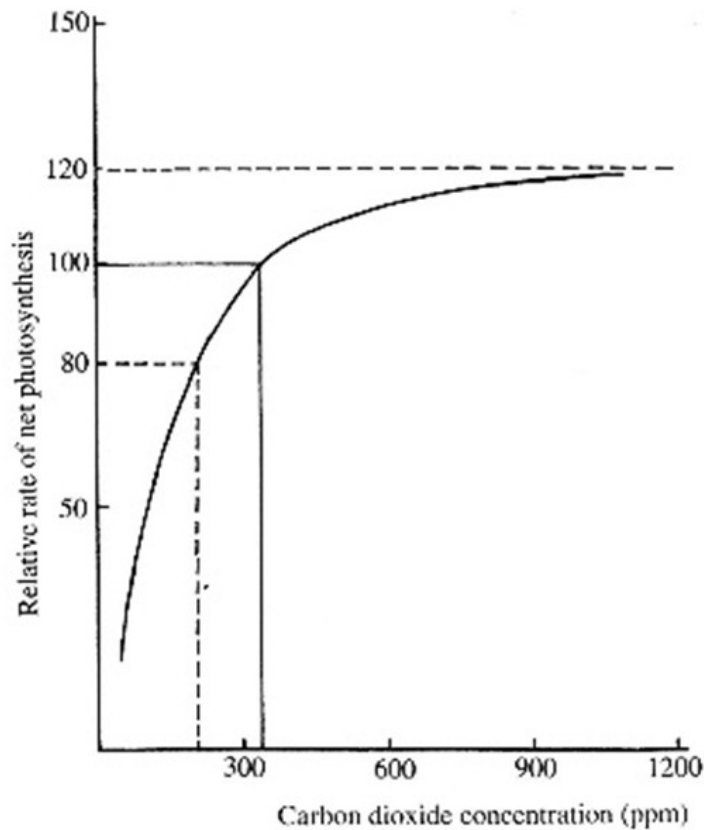


Figure 5.1: Effect of CO₂ concentration on photosynthesis rate. (UC Agriculture and Natural Resources, n.d.).

through simple ventilation can be sufficient to support healthy plant growth, without the need for costly CO₂ enrichment.

5.1.5 Water

A continuous supply of high-quality water is essential for plant transpiration, a process that not only cools the leaves but also facilitates the transport of nutrients from the roots to the leaves and fruits (Olabisi *et al.*, 2022).

To monitor soil moisture effectively, **Volumetric Water Content (VWC)** is commonly used. VWC is expressed as a percentage and represents the volume of water present in a given volume of soil.

$$\text{VWC (\%)} = \frac{\text{Volume of Water (cm}^3\text{)}}{\text{Volume of Soil (cm}^3\text{)}} \times 100 \quad (5.2)$$

This metric provides valuable information for irrigation management, ensuring that plants receive adequate moisture for optimal growth without the risk of overwatering or water stress.

5.2 Structural Design Standards

5.2.1 Architecture of a greenhouse

Beyond structural integrity, the architectural design of the greenhouse plays a key role in regulating its internal climate. Greenhouses function either to retain heat during cold periods or to exclude heat in hotter seasons, with the aim of maintaining a productive microclimate for plant growth (Ghani *et al.*, 2019). In South Africa's hot summers, a primary design consideration is minimizing heat buildup, making passive and active cooling essential.

One of the most influential architectural elements is the shape of the greenhouse, particularly the roof profile. The geometry directly affects solar irradiance levels inside the structure. In hot, arid environments, the optimal greenhouse form is one that minimizes solar gain during summer while maximizing it in winter, thereby supporting year-round cultivation. Research has shown that the roof angle and shape significantly influence the amount and distribution of solar radiation received (Ghani *et al.*, 2019).

A well-designed greenhouse structure must therefore strike a balance between mechanical resilience (to environmental loads) and thermal efficiency (to manage light and heat), with the shape and orientation of the building serving both functions. There are generally 4 main shapes discussed:

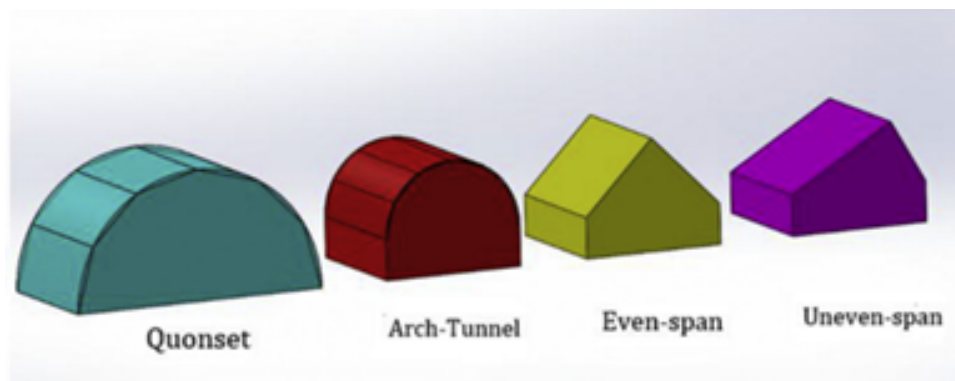


Figure 5.2: Main Greenhouse Shapes. (Sethi, 2009).

Choosing the right shape involves balancing structural load requirements with climate control goals, and often depends on regional weather patterns, crop types, and energy availability.

In terms of geographic suitability, greenhouse performance varies significantly with latitude and shape. For example, at 31°N latitude, even-span and Quonset greenhouses tend to receive more solar radiation in winter and less in summer, which is ideal for regions with significant seasonal temperature variation. However, at higher latitudes such as 50°N, uneven-span greenhouses are more effective for maintaining suitable internal conditions throughout the year (Sethi, 2009)

The inclination angle significantly affects solar heat gain by altering the amount of radiation absorbed by the greenhouse surface. As shown in the figure below, solar energy gaining rates decrease with steeper angles, particularly for the N-S orientation. While both orientations perform similarly at 17°, the E-W orientation consistently captures more en-

ergy at higher inclinations. This highlights the importance of optimizing both orientation and tilt to balance heat gain and internal climate control.

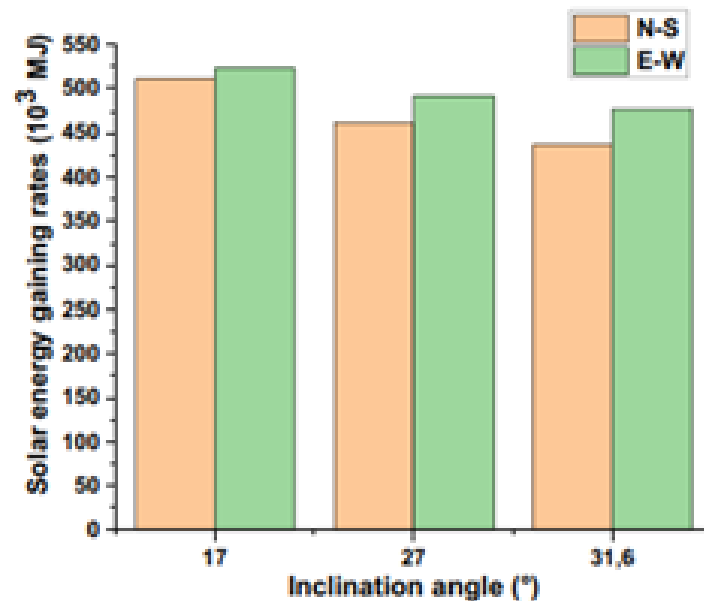


Figure 5.3: Solar radiation gain vs inclination angles (Mellalou *et al.*, 2022).

5.2.2 Sawtooth Design

The sawtooth greenhouse has gained popularity in tropical and subtropical climates due to its unique roof design and natural ventilation system. This roof features several curved, sawtooth-shaped sections, with vertical windows at the highest points to enhance natural ventilation. It uses natural resources like airflow and gravity to regulate the temperature and humidity inside the greenhouse, reducing reliance on external energy sources such as electricity (Insogreen, n.d.). Compared to traditional greenhouses, sawtooth greenhouses provide better ventilation, temperature control, and structural stability, making them ideal for hot, humid regions and areas with frequent rain. As the temperature inside the greenhouse rises, warm air naturally ascends and exits through the vertical windows at the top, while cooler air enters from below, creating a natural airflow cycle. The vertical windows are often controlled by a roll-up mechanism that opens for ventilation and closes for insulation when needed.

5.2.3 Construction techniques

Greenhouses are typically classified by structural type and the materials used for framing and covering. The primary structural types include wooden-framed, pipe-framed, and truss-framed greenhouses, each suited to different span widths and construction needs.

Wooden-Framed Structures: Ideal for spans under 6 meters, these greenhouses use timber, often pine or other locally sourced lumber, for columns and side posts. They are cost-effective and reasonably durable for small-scale applications and can also utilize bamboo as an eco-friendly alternative.

Pipe-Framed Structures: Used for medium spans up to approximately 12 meters, these



Figure 5.4: Sawtooth Greenhouses (Insogreen, n.d.).

structures incorporate metal pipes to form side posts, columns, cross ties, and purlins. Trusses are generally not included in this design.

Truss-Framed Structures: Suitable for larger spans of 15 meters or more, truss-framed greenhouses offer high structural strength and are commonly used for glasshouse construction. Trusses consist of welded elements like rafters, chords, and struts, made from flat steel, tubular steel, or angle iron. These frames are well-suited for pre-fabrication and large-scale installations.

Framing Materials: Common framing materials include wood, steel, and aluminium. While wood is inexpensive and accessible, aluminium is preferred for its strength, low maintenance, and resistance to corrosion, making it the most durable and cost-effective option for long-term greenhouse use (Dalai, 2020).

5.2.4 Covering materials

The choice of covering material greatly influences the internal environment, light transmission, durability, and cost of a greenhouse. The most used materials include glass, plastic film, and rigid panels.

Glass Greenhouses: Glass offers high light transmission, good air circulation, and lower internal humidity, which helps reduce disease incidence. However, they are more expensive to build and maintain compared to plastic-covered structures. They introduce additional maintenance challenges and potential issues due to glass's fragile nature.

Plastic Film Greenhouses: Covered with flexible materials such as polyethylene, polyvinyl chloride, or polyester, plastic film greenhouses are cost-effective and require less heating. Their main disadvantage is a shorter lifespan, even UV-stabilized films typically last no more than 4 years.

Rigid Panel Greenhouses: Made from polycarbonate, fiberglass-reinforced plastic, acrylic, or PVC rigid panels, these coverings offer better durability (up to 20 years) and more uniform light distribution. However, the panels are prone to collecting dust and algae, which

reduces light transmission over time (Verma *et al.*, 2022).

5.3 Enviromental control

Automated control of environmental parameters such as irrigation, heating and cooling, humidity, and airflow reduces manual workload and contributes to improved plant health. The following sections outline key industry standards and practices for managing these environmental factors effectively.

5.3.1 Irrigation Systems

The right water system is key to the success of any greenhouse project. As greenhouse plants have a limited soil buffer, precision irrigation systems are vital to ensure high-quality crops and yields (Vegtech Netafim, n.d.). Some of the most used types of irrigation include:

Drip Irrigation: This system delivers water directly to the base of each plant through a network of tubes and emitters, ensuring precise watering and minimizing water waste. By keeping foliage dry, it reduces the risk of fungal diseases. Drip irrigation is highly efficient and can be automated for consistent application (Cedar-Built Greenhouses, n.d.).

Overhead Sprinkler Systems: Overhead sprinklers distribute water across the greenhouse from above, providing broad coverage suitable for larger areas. While effective for uniform watering, they may increase humidity levels and wet plant foliage, potentially elevating the risk of disease. Proper management is essential to mitigate these concerns.

Misting Systems: Misting systems emit a fine mist of water into the air, increasing humidity and gently hydrating plants. They are particularly beneficial for delicate seedlings, tropical plants, or crops that thrive in high-humidity environments. Misting helps prevent over-saturation and promotes better air circulation.

Ebb and Flow (Flood and Drain) Systems: Also known as flood and drain systems, ebb and flow setups periodically flood the growing area with nutrient-rich water and then allow it to drain away. This cyclical process ensures that plant roots receive adequate moisture and nutrients while preventing waterlogging. Ebb and flow systems are commonly used in hydroponic greenhouse operations (FloraFlex Media, n.d.).

Irrigation water in these systems is most effectively stored in closed storage tanks. This provides a means to better control water quality as it protects water from outside contaminants, in addition to minimizing loss of water due to evaporation.

5.3.2 Heating and Cooling Systems

Heating and Cooling Systems in Greenhouses

Efficient heating and cooling systems are essential for maintaining optimal conditions within greenhouses. These systems can be broadly categorized into **passive** and **active** methods, each with distinct mechanisms and applications (Vegtech Netafim, n.d.).

Passive systems utilize natural processes and strategic design to regulate the greenhouse environment without relying on external energy sources, while **active systems** involve mechanical or electrical components that require external energy inputs.

Passive Heating

This approach leverages solar energy to naturally warm the greenhouse. By incorporating design elements such as proper orientation, insulation, and thermal mass (e.g., water barrels or rock walls), heat is absorbed during the day and released at night, maintaining a stable internal temperature (Kumari *et al.*, 2007).

Active Heating

To compensate for temperature drops, especially during winter nights, various heating systems are utilized. Options include unit heaters, central heating systems, radiant heat, and solar heating systems. The choice of system depends on factors such as greenhouse size, crop requirements, and energy efficiency considerations.

Passive Cooling

Natural ventilation is employed to facilitate air exchange between the interior and exterior, helping to dissipate excess heat. This can be achieved through operable vents in the roof and sidewalls, allowing warm air to escape and cooler air to enter (Faridi *et al.*).

Active Cooling

In situations where passive cooling is insufficient, active methods like evaporative cooling are implemented. This technique involves using fans and wet pads to cool the air through water evaporation, effectively reducing internal temperatures even during hot periods (Verma, 2022).

Integrating both passive and active systems allows for a comprehensive approach to climate control in greenhouses, ensuring optimal growing conditions across varying external temperatures and weather conditions.

5.3.3 Humidity Control

A cost-effective way to achieve humidity control in a controlled environment involves a misting system that sprays mist over the entire crop, thereby increasing humidity (Bluelab, n.d.).

It is essential to recognize that relative humidity is strongly influenced by temperature: warm air can hold more moisture than cooler air. As temperatures drop, the relative humidity increases—even if the actual amount of water vapor in the air remains constant.

Traditional humidity control methods often involve a combination of heating and ventilation. When humidity exceeds the desired level, greenhouse vents and windows can be opened to expel warm, moist air and allow cooler, drier air to enter. While this effectively reduces humidity, it can also result in unwanted heat loss. As a result, additional heating

may be required to maintain optimal growing temperatures. Achieving a balance between ventilation and heating is therefore essential for stable humidity and temperature regulation within the greenhouse environment (Drygair Greenhouse Dehumidifiers, n.d.).

5.3.4 Airflow

Airflow in Greenhouse Environments

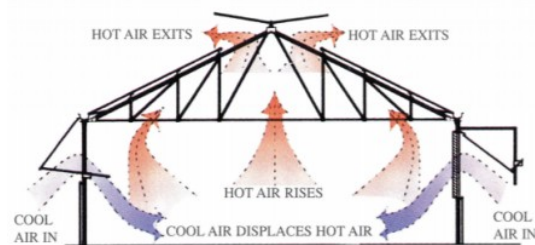
Effective airflow is crucial in greenhouse environments to maintain optimal growing conditions and prevent plant diseases. Proper air movement helps disperse microclimates, preventing localized areas of trapped moisture, and ensures uniform temperature and humidity levels throughout the greenhouse. This uniformity is vital for consistent plant growth and reducing the risk of fungal infections (Drygair Greenhouse Dehumidifiers, n.d.).

Airflow Components in Greenhouses

- **Roof Vents:** These allow hot air to escape from the top of the greenhouse, facilitating natural convection. As warm air rises and exits, cooler air is drawn in, aiding in temperature regulation.
- **Side Vents:** Positioned along the sides, these vents promote cross-ventilation by enabling fresh air to flow horizontally through the structure. This movement helps maintain consistent temperature and humidity levels.
- **Intake and Exhaust Fans:** Mechanical systems that actively move air in and out of the greenhouse. Intake fans draw in fresh, usually cooler air, while exhaust fans expel stale, humid air, preventing the buildup of moisture and heat (GrowersNetwork, n.d.).
- **Horizontal Airflow (HAF) Fans:** These fans circulate air horizontally within the greenhouse, ensuring even distribution of temperature and humidity. By moving air consistently, HAF fans help eliminate microclimates and reduce condensation on plant surfaces, thereby lowering the risk of disease (López *et al.*, 2013).



(a) Greenhouse profile



(b) Ventilation airflow

Figure 5.5: Airflow (GrowersNetwork, n.d.)

The figures above show an active ventilation system displaying the use of side and roof vents to promote healthy air circulation and thermal control. Integrating these compo-

nents effectively creates a balanced airflow system that supports healthy plant development by regulating environmental conditions and minimizing disease-promoting factors.

5.3.5 Automation and Environmental Control

Automation in greenhouses allows for precise control of environmental conditions such as temperature, humidity, light, and CO₂. Using sensors and controllers, these systems enhance efficiency, minimize manual labor, and support consistent crop development. Depending on the greenhouse's complexity, control systems can range from simple microcontrollers to advanced PLCs and IoT-based architectures (Farooq *et al.*, 2022).

5.3.5.1 Sensors

- **Temperature Sensors:** Monitor the internal temperature of the greenhouse.
Common types: Thermistors, RTDs, DS18B20 (digital)
- **Humidity Sensors:** Measure the ambient humidity levels.
Common types: DHT22, SHT31
- **Soil Moisture Sensors:** Detect the moisture content in soil.
Common types: Capacitive soil moisture sensors, resistive sensors
- **Light Sensors:** Measure light intensity reaching plant surfaces.
Common types: BH1750, TSL2561, TSL2591
- **CO₂ Sensors:** Monitor carbon dioxide concentration.
Common types: MH-Z19, Senseair S8

Sensor data is fed into a controller, such as a PLC, Arduino, or Raspberry Pi, enabling automated decision-making for heating, cooling, irrigation, and other critical operations (WebbyLab, n.d.a).

5.3.5.2 Controllers

Programmable Logic Controllers (PLC): A Programmable Logic Controller (PLC) is a robust, industrial-grade computer used for controlling and monitoring industrial processes. It operates by receiving inputs, processing the data, and triggering outputs based on pre-programmed logic. PLCs are widely used in greenhouse automation due to their flexibility and reliability. Built with rugged designs and solid-state components, PLCs ensure long-term, reliable performance in harsh environments. The main drawback of PLCs is their higher power requirements and higher cost compared to microcontroller-based systems, which may limit their use in smaller or cost-sensitive projects (Dwinugroho *et al.*, 2021).

Microcontrollers: Microcontrollers are small, programmable devices that can control a wide range of greenhouse automation tasks, such as environmental monitoring and data logging. The following microcontrollers are commonly used in greenhouse automation:

- **Raspberry Pi:** A low-cost, powerful computer board frequently used in Internet of Things (IoT) applications. It offers greater processing power and versatility than simpler microcontrollers like Arduino and ESP8266. Raspberry Pi is ideal for more complex greenhouse automation systems, including advanced data collection and analysis (Raspberry Pi Foundation, n.d.).

- **ESP8266/ESP32:** These low-cost Wi-Fi microchips, developed by Espressif Systems, include built-in TCP/IP stacks and microcontroller capabilities. The ESP8266 and ESP32 are widely used in IoT applications for connecting devices to the internet. They are compatible with the Arduino IDE, making them a popular choice for developing cost-effective, connected greenhouse systems (Espressif Systems, n.d.).
- **Arduino:** An open-source platform based on simple microcontroller chips, Arduino is designed for ease of use and flexibility. It is suitable for basic greenhouse monitoring and control tasks. Arduino's simplicity and large community make it an excellent choice for entry-level projects that require straightforward automation (Yahaya *et al.*, 2019).

These controllers play a critical role in facilitating data logging and remote monitoring as well as enabling automated greenhouse systems to efficiently manage environmental factors such as temperature, humidity, and light for improved crop management.

Wireless Connectivity and Actuation

Wireless Connection:

One can make use of ThingSpeak with both Raspberry Pi and Arduino to wirelessly upload and visualize sensor data. ThingSpeak is an IoT analytics platform that allows real-time data collection, storage, and visualization from connected devices. Once connected to Wi-Fi, the Arduino or Raspberry Pi can send HTTP requests to ThingSpeak's API to upload sensor readings. By integrating ThingSpeak with these controllers, one can effectively monitor and analyze sensor data remotely, making it a valuable tool for greenhouse automation and other IoT applications (MathWorks, n.d.).

Actuators and Switches:

In greenhouse automation, relays are essential components that allow low-voltage control signals to switch high-voltage appliances such as irrigation pumps, lights, and ventilation systems on and off. This ensures efficient and safe operation of the system. These relays can be activated using the low voltage signals sent by the controller (WebbyLab, n.d.b).

5.4 Energy Efficiency & Sustainability

Solar power is useful in greenhouses because it provides a renewable and sustainable energy source for powering equipment, leading to reduced operating costs and environmental impact, while also enabling year-round crop production. By harnessing solar energy, solar-powered greenhouses create sustainable growing conditions for plants, regardless of external climate variations. With a system that runs on the solar power, especially when paired with a battery, it is not necessary to rely on an energy provider or worry about blackouts.

5.4.1 Key Requirements for Solar System

Incorporating a solar power system into a greenhouse requires understanding its essential components, each playing a pivotal role in energy capture, conversion, storage, and utilization. The key elements include:

1. **Solar Panels (Photovoltaic Modules):** Solar panels are the primary components that capture sunlight and convert it into direct current (DC) electricity through photovoltaic cells (Krisciunas, 2023).
2. **Inverter:** The inverter transforms the DC electricity generated by the solar panels into alternating current (AC) electricity, which is the standard form used to power electrical devices and systems within most greenhouses.
3. **Charge Controller with Maximum Power Point Tracking (MPPT):** A charge controller manages the flow of electricity from the solar panels to the batteries, preventing overcharging and prolonging battery life. MPPT or Maximum Power Point Tracking is an algorithm included in charge controllers used for extracting maximum available power from a PV module under varying conditions (Eltawil and Zhao, 2013).
4. **Batteries:** Batteries store the electricity produced by the solar panels, providing energy during periods without sunlight or during power outages, ensuring continuous operation of greenhouse systems.
5. **Racking and Mounting Systems:** Proper racking and mounting systems are crucial for securely installing solar panels on the greenhouse structure or the ground. These systems ensure that panels are positioned at an optimal angle and orientation to maximize solar exposure. Standard rails and brackets for panels of various sizes exist on the market (Axe Struct, n.d.).

Understanding and integrating these components effectively is vital for the successful implementation of a solar-powered greenhouse, leading to enhanced energy efficiency and sustainability.

6 Concept Design

This section outlines the concept development for the electronic greenhouse. It aims to minimise concept vulnerability by considering many concepts, both good and bad, before choosing one concept that is the best fit to fulfil the desired role.

6.1 Problem Definition

Greenhouses require continuous supervision to maintain optimal growing conditions for plants. This manual oversight can be time-consuming and inconsistent, potentially affecting crop quality and yield. The proposed solution is to develop an electronic greenhouse system capable of monitoring and controlling critical environmental parameters such as temperature, humidity, soil moisture, CO₂ levels, and light intensity. The design should support plant growth while minimizing human intervention. This project focuses on designing and developing a small-scale, automated greenhouse system suitable for educational or personal use. The scope includes:

- Selection and integration of sensors for monitoring environmental variables (e.g., temperature, humidity, CO₂, light).
- Design and implementation of a control system using microcontrollers or PLCs.
- Incorporation of solar power as a primary energy source.
- Design of a structural frame using lightweight, transparent materials.
- Implementation of programmable irrigation and ventilation systems.
- Development of a simple user interface or data display.
- Testing and documentation of system performance under typical operating conditions.

The project will not cover large-scale commercial greenhouse operations or complex AI-driven automation but will be designed to allow future scalability or modular upgrades.

6.2 Stakeholder Requirements

Table 6.1 expands on the problem definition given above. The items in the table represent the requirements as stated by the stakeholders. The stakeholders in this project are the end-user, the University of Stellenbosch staff and supervisors, and health and safety officers in compliance with the OSHA Act. The end-user benefits from the smooth operation of the final product, while both the end-user and the university benefit from the design being the most cost-effective option, given that university funds are limited.

Table 6.1: Stakeholder Requirements

Number	Stakeholder	Description	Priority
SR1	End-user	Secure frame and covering material creating thermally insulated space for environment control.	Must-have
SR2	End-user	Monitor environment.	Must-have
SR3	End-user	Data logging capabilities for analysis and decision making.	Must-have
SR4	End-user	Actuation allowing for control of ventilation, air-flow, temperature, lighting, and humidity.	Must-have
SR5	End-user	Irrigation system allowing for control of soil moisture content and humidity.	Must-have
SR6	End-user	System powered by solar energy.	Must-have
SR7	End-user, University	Cost-effective.	High
SR8	End-user, University	Energy efficient.	High
SR9	End-user	Environmental parameter allowable ranges settable by user with simple user interface.	High
SR10	University	Easy to manufacture.	High
SR11	OSHA	Comply with safety standards.	Must-have

6.3 Engineering Requirements

Table 6.2: Functional Requirements

FR Number	Description	Related Stakeholder Requirement
1	Allow transmittance of sunlight	SR1
2	Monitor temperature ¹	SR2
3	Monitor soil moisture	SR2
4	Monitor humidity ¹	SR2
5	Monitor CO ₂	SR2
6	Monitor light levels	SR2
7	Display environment conditions	SR3, SR9
8	Process environment conditions	SR3
9	Control temperature	SR4
10	Control humidity	SR4, SR5
11	Control ventilation	SR4
12	Control airflow	SR4
13	Control irrigation	SR5
14	Supply light	SR4
15	Withstand weather conditions	SR1
16	Power the system	SR6, SR8

¹ Refers to both internal and external temperature and humidity.

Table 6.3: Performance Requirements

PR Number	Description	Target	Range	Unit	Related Stakeholder Requirement
1	Thermal Insulation	5	>5	$\text{m}^2 \cdot \text{K/W}$	SR1
2	Resist Wind speeds ¹	18.81	>18.8	kph	SR1
3	Temperature sensing range	Maximize	-10-50	°C	SR2
4	Soil moisture sensing	0-100	0-100	%	SR2
5	Humidity sensing	0-100	0-100	%	SR2
6	CO2 sensing range	0-1000	0-1000	ppm	SR2
7	Light sensing ²	100-600	0-1000	$\mu\text{mol} \cdot \text{m}^{-2} \cdot \text{s}^{-1}$	SR2
8	Latency in displaying environmental conditions	Minimize	<120	s	SR3
9	Control temperature ³	User selected	15-30	°C	SR4
10	Control soil moisture content	User selected	0-100	%	SR5
11	Control humidity ⁴	User selected	40-80	%	SR5
12	Supplemental lighting intensity ²	300	200-400	$\mu\text{mol} \cdot \text{m}^{-2} \cdot \text{s}^{-1}$	SR4
13	Energy Consumption	Minimize	0-1200	Wh	SR8
14	Battery storage capacity	48	>48	hours	SR6

¹ Stellenbosch Climate – Weather Spark, n.d. Wind speeds peak at 18.8 kph in Jan.

² A Definitive Grow Light Study, n.d. Seedlings: 100-300, Vegetative: 300-600 $\mu\text{mol} \cdot \text{m}^{-2} \cdot \text{s}^{-1}$

³ Koşan et al., 2020. Ideal temperature: 15-30°C.

⁴ Ruuvi, n.d. Recommends 40-80% humidity for greenhouses.

6.4 Functional Decomposition

The functional requirements outlined in the previous section describe only the high-level actions that the system must perform, as perceived by stakeholders. In this section, these functions are decomposed to identify the internal operations that the system must carry out to meet those requirements.

Figure 6.1 illustrates the flow of energy, information, and/or materials into and out of the system's internal functions. These functions are responsible for consuming, generating, or transforming the various flows to support overall system operation.

The primary subsystems and their corresponding functions are numbered from 1.0 to 3.0.

These serve as the foundation for the physical decomposition discussed after the concept evaluation.

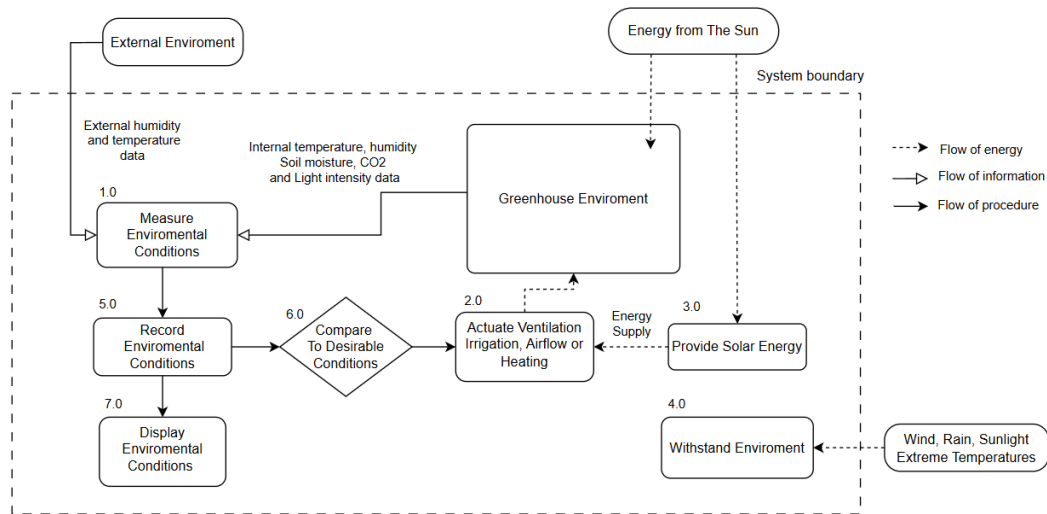


Figure 6.1: Functional decomposition

6.5 Concept Development

To address the multidisciplinary requirements of the concept design, the system will be partitioned into specialized subsystems. Each subsystem concept is derived from comprehensive market research, focusing on industry-standard operations and adapting these practices to suit the scale and specific needs of a small-scale greenhouse environment. Each subsystem will undergo an independent evaluation to identify solutions that optimally balance functional performance, efficiency, and cost-effectiveness. By analyzing multiple design concepts for every subsystem, this approach aims to mitigate the risk of concept vulnerability. This methodology aligns with established systems engineering practices, where alternative designs are explored and analyzed to optimize the overall system architecture.

6.6 Concept Evaluation

In this section the developed concepts are evaluated and expanded on. A weighted scoring system was used to evaluate the best design for each subsystem. Key criteria like efficiency, cost and manufacturability were assigned importance to weights. Each concept was rated 1-5 against these criteria, with scores calculated by multiplying ratings by weights and summing the results. The highest-scoring concepts were selected, ensuring optimal performance while meeting project priorities. This approach minimized bias and enabled clear trade-off decisions between competing requirements.

6.6.1 Structure

For the structure there are generally 3 options that will fit with the small-scale greenhouse's requirements. The criterion are weighted by a percentage and the concepts are rated on a scale of 1-5. The weights were selected to reflect the priorities for a small-scale greenhouse: temperature management is most critical (30%), followed by cost effectiveness (30%), while manufacturability (20%) and space efficiency (20%) are also key considerations.

Criterion	Weight (%)	Even Span	Lean-to	Arch	Rationale
Temperature Management	30	3	3	3	All options allow moderate control; lean-to benefits from thermal mass of shared wall, but may overheat in summer.
Ease of Manufacturability	20	2	4	1	Lean-to can require fewer materials by leveraging an existing wall; even span is moderate; arch is most complex.
Space Efficiency & Footprint	20	2	3	2	Lean-to can optimize space against a building; others require more free-standing footprint.
Cost Effectiveness	30	2	3	1	Lean-to reduces material and covering costs; even span moderate; arch requires more structure and covering.
Weighted Total	100	2.3	3.2	1.8	Lean-to achieves the best balance across cost, space, and manufacturability.

Table 6.4: Weighted Concept Evaluation of Greenhouse Structures

For small-scale applications, the lean-to option emerges as the most practical solution, balancing cost, space utilization, and manufacturability, while still offering adequate thermal performance.

6.6.2 Covering Material

The weighting was chosen to reflect the relative importance of each criterion in greenhouse performance. Durability (30%) was weighted highest as the covering must withstand weathering and mechanical stress. Light transmissibility (20%) and thermal insulation (20%) were both critical to plant growth. Cost effectiveness (20%) was included to balance performance with budget, while ease of manufacturability (10%) was given a smaller weight since fabrication complexity is less important than long-term functionality.

Criterion	Weight (%)	Glass	Soft Plastic Covering	Perspex Acrylic	Rationale
Light Transmissibility	20	4	3	3	Glass transmits light exceptionally well; soft plastic is less transparent; Perspex is good but typically slightly lower than glass.
Durability	30	2	1	4	Glass is brittle and prone to breakage; soft plastic tears/UV-degrades; Perspex is impact-resistant with good weathering.
Cost Effectiveness	20	3	5	2	Soft plastic has lowest upfront cost; glass moderate; Perspex is higher cost per m ² .
Ease of Manufacturability	10	1	2	3	Glass is heavy and difficult to cut/fit; soft plastic easier to handle; Perspex is lightweight and easy to cut/shape.
Thermal Insulation	20	3	1	3	Glass and Perspex provide reasonable insulation; thin soft plastic performs poorly.
Weighted Total	100	2.7	2.3	3.1	

Table 6.5: Weighted Concept Evaluation of Greenhouse Covering Materials

Glass is a common greenhouse material providing good thermal insulation and light transmissibility. However, compared to glasses brittle nature Perspex is easier to cut and shape as convenient. Perspex is also more durable than soft plastic coverings and will not have to be replaced as often due to rips and tears that the soft plastic is prone to. Additionally, Perspex provides good thermal insulation and high light transmittance, making it a practical and cost-effective choice for greenhouse applications.

6.6.3 Active Heating

Temperature management and cost effectiveness were weighted highest at 30% each, as maintaining a stable climate and operating affordability are the most critical factors for long-term success.

Table 6.6: Weighted Concept Evaluation of Heating Options

Criterion	Weight (%)	Electric Heater	Gas Heater	Heat Lamp	Rationale
Temperature Management	30	3	4	3	Gas provides strong heating; electric moderate; lamps uneven.
Ease of Manufacturability	20	4	1	2	Electric is plug-and-play; gas requires plumbing; lamps simple but less effective.
Off-Grid Capabilities	20	4	1	1	Electric integrates with solar; gas not sustainable; lamps not suited for solar.
Cost Effectiveness	30	2	3	1	Gas fuel is cheaper; electric moderate; lamps inefficient per area.
Weighted Total	100	3.1	2.5	2.2	

While gas heaters can also elevate CO₂ levels, their integration adds cost and complexity to the greenhouse system, making them less suitable for the intended design. An electric heater is selected for its adaptability to varying heating requirements and its compatibility with the solar power system, enabling off-grid operation. Heat lamps tend to concentrate heat in localized areas rather than distributing it evenly throughout the greenhouse, potentially leading to inconsistent temperature regulation and increased energy consumption.

6.6.4 Active Cooling

The weighting was selected to reflect the relative importance of cooling performance in maintaining plant health. Temperature management (30%) was given the highest weight as this is the primary purpose of cooling. Cost effectiveness (30%) was also heavily weighted due to budget constraints for long-term operation. Off-grid capabilities (20%) were prioritized to ensure compatibility with renewable systems. Finally, ease of manufacturability (20%) was included to account for installation complexity and practicality.

Criterion	Weight (%)	Actuated Ventilation	Evaporative Cooling Module	Misting	Rationale
Temperature Management	30	4	2	3	Ventilation exchanges hot air for ambient; evaporative modules underperform in humidity; misting adds moderate evaporative cooling.
Ease of Manufacturability	20	4	1	3	Fans + simple controls vs. pumps, pads, and ducting; misting needs basic plumbing and nozzle layout.
Off-Grid Capabilities	20	3	2	2	DC/efficient fans suit solar; evaporative modules draw more power; misting depends on reliable pressurized water.
Cost Effectiveness	30	3	1	3	Ventilation: moderate capex/opex; evaporative: high capex and energy; misting: low capex but ongoing water use.
Weighted Total	100	3.5	1.5	2.8	

Table 6.7: Weighted Concept Evaluation of Cooling Options

Actuated ventilation emerges as the most economical and energy-efficient cooling method, as it requires only fans to exchange hot and cool air. Its drawback is that it can only cool the greenhouse down to the ambient temperature outside. To enhance this, active ventilation can be combined with misting. Misting promotes evaporative cooling within the greenhouse and, although it requires a pressurized water supply and humidity control, it significantly improves temperature regulation. In contrast, evaporative cooling modules are less favorable due to their high cost, greater power demand, and limited performance in humid environments. Therefore, a combined ventilation and misting approach is the most balanced solution, addressing both cost-effectiveness and cooling performance.

6.6.5 Irrigation

The weighting was selected to reflect the importance of reliable soil moisture control (30%) and cost effectiveness (30%) as primary design drivers. Coupled environment control capabilities (15%) were included to account for irrigation methods that are able to also influence temperature and humidity. Ease of manufacturability (15%) was considered for practicality, while off-grid capabilities (10%) were given a smaller weight as irrigation can often be supported by external water supplies.

Criterion	Weight (%)	Drip Fed	Ebb & Flow	Misting	Rationale
Soil Moisture Control	30	3	3	2	Drip and ebb-flow target roots well; misting less precise for soil.
Ease of Manufacturability	15	4	1	4	Drip/mist easy to set up; ebb-flow complex with pumps/reservoirs.
Off-Grid Capabilities	10	3	2	1	Drip can run on solar pumps; ebb-flow requires circulation; misting needs high pressure.
Coupled Env. Control	15	0	0	5	Only misting can also improve humidity/temperature Control.
Cost Effectiveness	30	4	2	4	Drip and misting affordable; ebb-flow expensive/complex.
Weighted Total	100	3.00	1.85	3.25	

Table 6.8: Weighted Concept Evaluation of Irrigation Methods

Misting emerges as the most effective method due to its dual role in both irrigation and environmental control, despite requiring higher water pressure. Drip irrigation remains economical and reliable for soil moisture but lacks climate control benefits. Ebb and flow systems provide uniform watering but is less practical due to cost and complexity. A misting system with sensor-based control is therefore preferred for optimizing both plant growth and climate regulation.

6.6.6 Controller Selection

The weights were assigned based on the relative importance of each criterion to the concept: GPIO flexibility (30%) is most critical for hardware interfacing, programmability and cost each carry significant weight (20% each), while WiFi (20%) and power consumption (10%) are also key but secondary factors. PLCs were considered but ruled out early due to their high cost, complexity, and limited suitability for rapid prototyping or application in small-scale systems.

Criterion	Weight (%)	Raspberry Pi 4B	ESP32	Arduino Uno	Rationale
Input/Output GPIO Flexibility	30	3	4	3	ESP32 has more versatile GPIO including analog inputs; Pi and Arduino provide sufficient but less flexible options.
Ease of Programmability	20	4	4	4	All three have strong community support and beginner-friendly development environments.
WiFi Capabilities	20	4	4	2	ESP32 and Pi include WiFi by default; Arduino requires shields/modules, increasing cost and complexity.
Power Consumption	10	4	3	3	Raspberry Pi is relatively efficient; ESP32 moderate; Arduino low but limited in capability.
Cost Effectiveness	20	2	4	3	ESP32 offers the best feature-to-price ratio; Arduino is low-cost but less capable; Raspberry Pi is more expensive.
Weighted Total	100	3.3	3.9	3.0	ESP32 emerges as the most balanced choice overall.

Table 6.9: Weighted Evaluation of Microcontroller Options

The ESP32 Offers a balance of performance and cost, with built-in Wi-Fi and Bluetooth, making it ideal for IoT and wireless applications.

6.7 Physical Decomposition

The following physical decomposition figures break down the physical components necessary in each subsystem while. The use of a system engineering approach delves into physical decomposition of the main subsystems (namely Environmental monitoring, Actuation, and Solar Power System) and are numbered 1 through 3 as seen in the functional decomposition in figure 6.1.

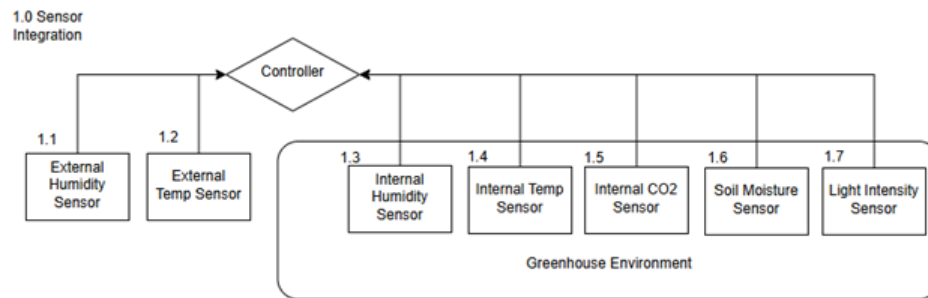


Figure 6.2: Sensor Physical Decomposition

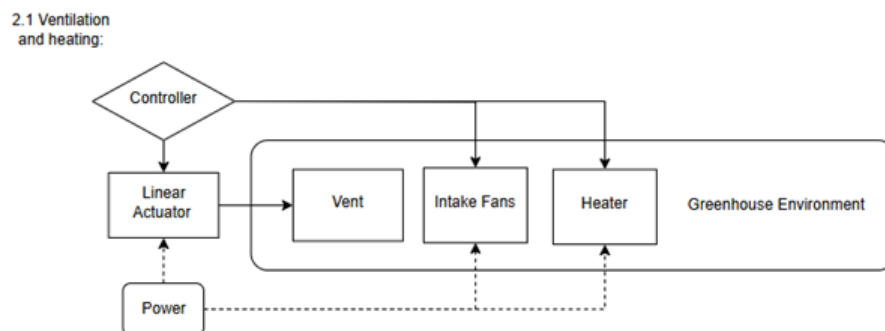


Figure 6.3: Ventilation Physical Decomposition

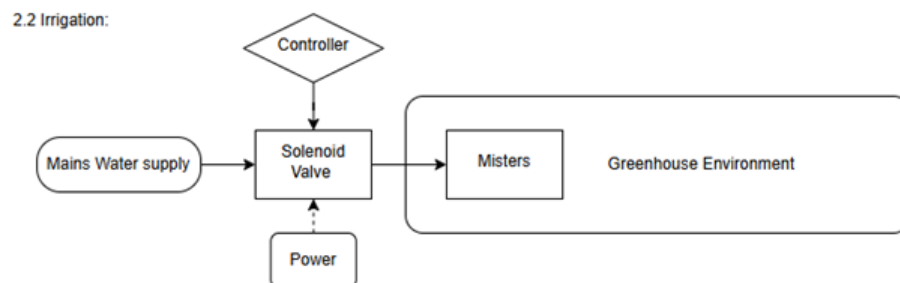


Figure 6.4: Irrigation Physical Decomposition

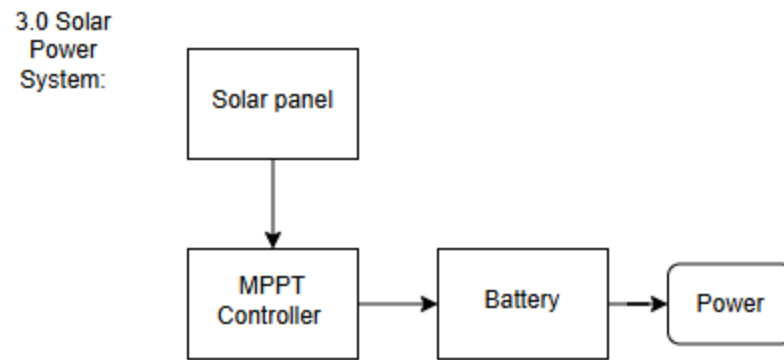


Figure 6.5: Solar Physical Decomposition

7 Detailed Design

Following a design review, a revised structural design was developed to enable improved operation and manufacturability. The general shape was optimized to accommodate laser cutting processes, with the maximum available bed size of 1260 mm × 860 mm. The roof geometry was modified to an uneven-span lean-to configuration. While this maintains a steep roof inclination for efficient solar gain, it also allows for better thermal control and ensures that the vent can be actuated even when the greenhouse is positioned adjacent to a wall.

Allowing users to place the greenhouse against a thermally massive surface (e.g., a concrete or brick wall) enhances passive heat retention and improves overall energy performance. Additionally, the revised structure introduces more right angles, simplifying manufacturing and increasing the usable internal volume of the greenhouse.



(a) Frame Structure



(b) Final Concept

Figure 7.1: Detailed Design

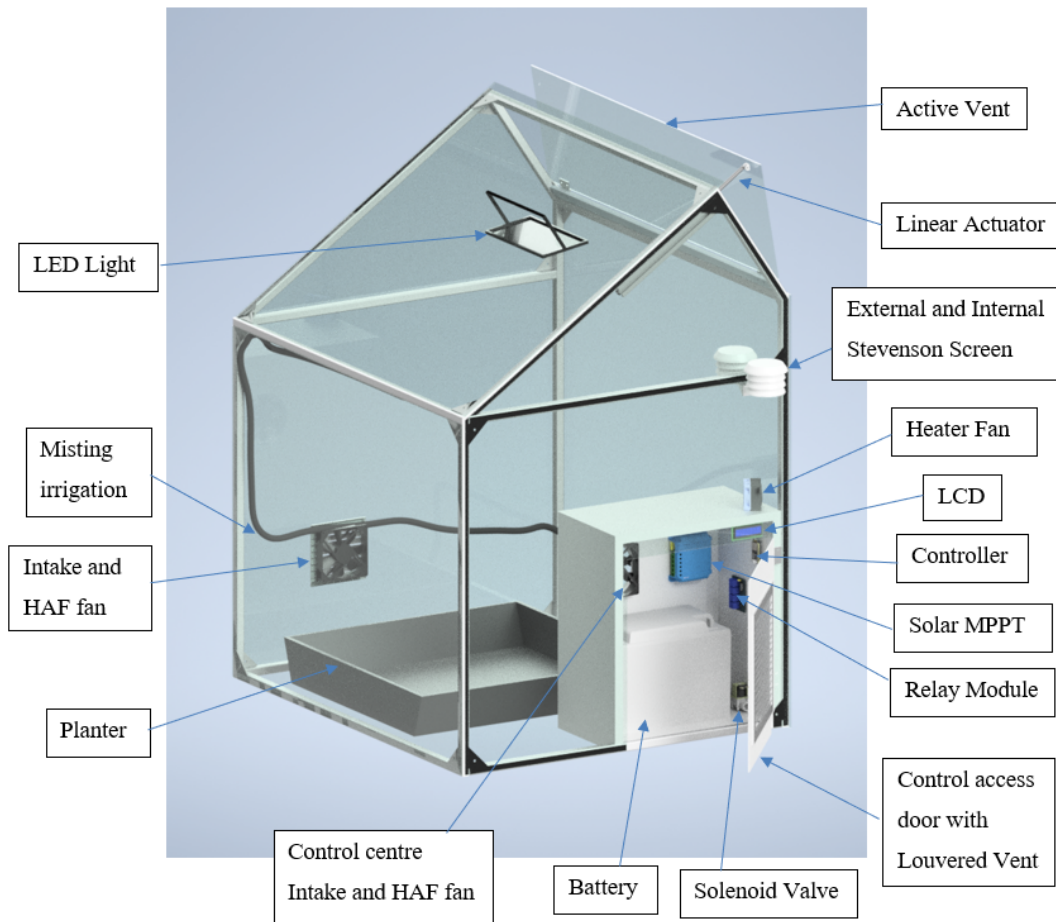


Figure 7.2: Final Concept Definition

The control center protecting all of the electrical components is shifted to the side of the greenhouse and a door is laser cut from the right side Perspex panel allowing easy access for maintenance. This side access preserves the ability to have the greenhouse be leveraged against an existing wall maximizing space efficiency. Hinges on the front Perspex panel enable it to open for convenient access required for planting and harvesting.

7.1 Structure

7.1.1 Frame

Aluminum square tubing is selected for its excellent strength-to-weight ratio. Compared to a stainless steel frame, aluminum offers a much lighter structure, making the greenhouse easier to transport or reposition if necessary. Additionally, aluminum is naturally corrosion-resistant, eliminating the need for protective coatings to prevent rust. It is also widely available and relatively easy to cut and assemble, making it a practical and cost-effective choice for the frame. A structural analysis seen in the calculations in Appendix D confirmed that the frame can withstand its own weight, the weight of Perspex panels, and wind loads simultaneously. Under maximum wind speeds, the maximum deflection simulated at the apex of the greenhouse amounted to only 0.5 mm, demonstrating excellent structural stiffness and integrity.

7.1.2 Covering material

Perspex was chosen over glass for its combination of machinability, thermal performance, transparency, and non-brittle nature, addressing both construction and long-term durability needs.

7.2 Intake and HAF fans

Intake fans bring in fresh oxygen rich air when needed. To regulate the greenhouse temperature, cool air is drawn in from the base while a top vent opens to allow warm air to escape, promoting natural convection and effective cooling. Industrial greenhouses also rely on HAF fans to promote healthy air movement and prevent stagnation. However, on a small scale dedicated HAF fans are not necessary. For the sake of simple design and using smart controller logic these two actions of air intake and HAF can be combined to be completed by the same two components. In this case two 12V fans on either side of the greenhouse are placed towards the bottom section of the greenhouse. These two fans will allow for air intake when necessary and intermediate HAF at a set interval. The fans are placed slightly offset from each other to promote circular airflow and improve coverage.

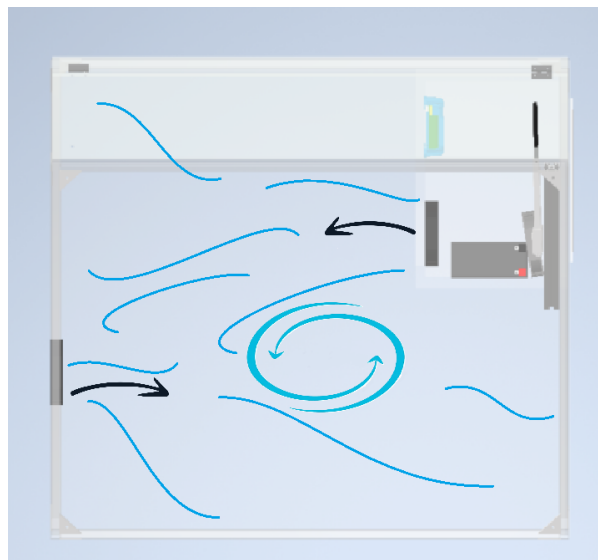


Figure 7.3: Air circulation

7.2.1 HAF Requirements

A common guideline is to size the HAF fans to be able to move 2 CFM for each square foot of greenhouse floor area. By these standards for the floor area of 0.85 m^2 an HAF capacity of at least 18.3 CFM would be required to ensure healthy air circulation. Two 120 mm PC fans with an average of 70 CFM each will be more than sufficient and will be activated during dedicated HAF periods.

$$A_{\text{floor}} := b \cdot l = 0.85 \text{ m}^2 \quad (7.1)$$

$$A_{\text{floor,ft}} := A_{\text{floor}} \cdot 10.764 \frac{\text{ft}^2}{\text{m}^2} = 9.1494 \text{ ft}^2 \quad (7.2)$$

$$CFM_{\text{HAF}} := A_{\text{floor,ft}} \cdot 2 \frac{\text{ft}}{\text{min}} = 18.2988 \frac{\text{ft}^3}{\text{min}} \quad (7.3)$$

Exhaust fans are not necessary as the vent at the top, having a large surface area, will allow for the escape of a large quantity of air when the air intake and ventilation is active. This exhaust flow is assisted by the buoyancy effect as stipulated in the calculations.

Due to the use of misting systems within the greenhouse care needs to be taken to protect the fans and other components from moisture buildup, for this reason 3D printed covers are designed to protect from droplet buildup on components. Smart placement of misting nozzles and intermediate misting control will also prevent excessive moisture exposure to electrical components.

7.3 Ventilation

7.3.1 Natural Ventilation

An effective ventilation system at the top of the greenhouse will allow for passive cooling using the buoyancy effect of the warm air within the greenhouse. The larger the area of the vent the more effective the air transfer and thus cooling capabilities. A Perspex panel on the roof measuring 1000mm by 270mm is to be actuated to the open position to allow for this exchange. A linear actuator is selected as it is less complicated than installing a motor and shaft and more cost effective. The actuator selected should have a large stroke of at least 100mm to increase the opening area and will need to be rated to lift the weight of the Perspex panel. Calculations were completed as seen below to ensure that the specification can be met.

Vent Actuation:

$$\rho_{\text{perspex}} := 1,19 \frac{\text{g}}{\text{cm}^3} \quad \text{density of perspex}$$

$$w_{\text{vent}} := 270 \text{ mm} \quad \text{width of vent}$$

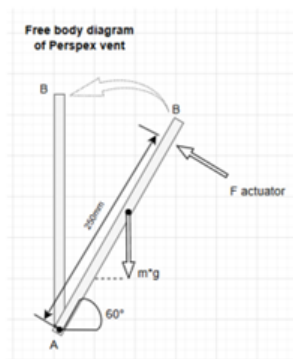
$$m_{\text{vent}} := b \cdot w_{\text{vent}} \cdot t \cdot \rho_{\text{perspex}} = 0,9832 \text{ kg} \quad \text{mass of vent}$$

$$\theta := 60^\circ \quad \text{angle of inclination of vent}$$

$$F_{\text{weight}} := m_{\text{vent}} \cdot 9,81 \frac{\text{m}}{\text{s}^2} = 9,645 \text{ N} \quad \text{weight of vent acting at centroid}$$

$$M_{\text{weight}} := (-F_{\text{weight}}) \cdot \frac{w_{\text{vent}}}{2} \cdot \cos(\theta) = 1,2401 \text{ N m} \quad \text{moment around point A due to weight}$$

$$F_{\text{actuator}} := \frac{M_{\text{weight}}}{0,25 \text{ m}} = 4,9604 \text{ N} \quad \text{assuming that the linear actuator is connected to the panel at 250mm from the hinges at A and that it is initially acting perpendicular to the panel}$$



When opened, the vent exposes the greenhouse interior to potential precipitation, which could result in overwatering or water exposure to electrical components. To mitigate this,

a rain sensor is mounted on the exterior of the greenhouse and connected to the microcontroller. When moisture is detected, the sensor signals the system to close the vent, thereby protecting both the plants and internal electrical components during periods of rain.

7.3.2 Forced Ventilation

Typical greenhouse design guidelines recommend aiming for one complete air exchange per minute. (Watson et al., 2019) with the greenhouse volume equating to 0.77m³ two fans would be required to have a flowrate of at least 0.77m³ /min

Fan Sizing:

$$V := b \cdot h_2 \cdot l - \frac{(h_2 - h_1)}{2} \cdot b \cdot l = 0,7735 \text{ m}^3 \quad \text{total volume of greenhouse}$$

$$flow := \frac{V}{\text{min}} = 0,7735 \frac{\text{m}^3}{\text{min}} \quad \text{total flow required by fans per minute}$$

$$CFM := flow \cdot 35 \frac{\text{ft}^3}{\text{m}^3} = 27,0725 \frac{\text{ft}^3}{\text{min}} \quad \text{total flow in CFM}$$

The combined volumetric flow-rate of intake fans would have to be at least 27 CFM. For this two PC fans are selected each averaging 70 CFM more than capable of transferring the entire volume of the greenhouse within a minute.

7.4 Misting System

Proper greenhouse management involves regulating soil moisture through watering while maintaining humidity levels within the ideal range for the cultivated plants. Instead of two separate systems complicating the design it is decided to implement a twin operation where a single misting system along with smart scheduling and programmed procedures can strike a balance between controlling the supply of fresh water as well as allowing for the control of the humidity in the environment. One way in which this can be implemented is that when watering is necessary during and after a subsequent misting period the ventilation system is activated to release excess humidity and drop back to the desired level. The added benefit of having a misting system is that it also allows for increased cooling capabilities and will work alongside the active and passive ventilation system to cool the greenhouse when required.

Misting Nozzles are placed effectively to ensure moisture reaches the roots directly. Up to 4 nozzles will be placed slightly above the soil and deliver water straight to the soil medium. Misting nozzles offer adjustable spray patterns, ranging from a targeted drip to an ultra-fine mist. Two additional nozzles will be installed at a higher elevation within the greenhouse to provide broader coverage while still targeting specific zones. Careful positioning will ensure mist is directed away from sensitive electrical components.

7.5 Active Heating System

To achieve effective active heating, a fan heater is selected. Based on the heat loss calculations on the surface area, the greenhouse experiences a convective heat loss of approximately 119 W when there is a 5 °C temperature difference between the internal and external environments.

Given that many plants require a minimum desirable temperature of around 15 °C for optimal growth, the heating system must be capable of compensating for this heat loss. Therefore, a fan heater with a power rating of at least 120 W is required.

A 12 V, 120 W fan heater is chosen to meet this requirement, powered directly by the 12 V battery supply. The heater is controlled via a relay module, which is switched by signals from the microcontroller. This setup allows the heater to automatically activate when the internal temperature falls below a user-defined threshold.

7.6 Control Center Design

7.6.1 Electrical Component Housing

A dedicated area is required for the safe and neat storage of all electrical components such as the battery, controller, and valve. The control center can be made from wood as it is simple to construct and mount parts to, as well as being electrically nonconductive and thermally insulating.

Specifically, marine plywood will be used as it resists water damage and is made from high-quality hardwood veneers, making it stronger and more resistant to warping compared to standard plywood. Coating it with white exterior-grade paint will further improve water resistance and reflect sunlight, keeping internal components cooler while reflecting more light for the plants.

7.6.2 Ventilation System

A cutout from the Perspex allows for placement of a louvre vent. These vents will allow for passive ventilation to keep the control center cool while the shape of the louvre vents will prevent water from entering. This vent needs to be placed on the side of the greenhouse to allow for the placement of the greenhouse against a wall.

For this reason, the control center was shifted to the right side of the greenhouse, which also allows for more space to be used for the plants. The system incorporates dedicated ventilation to prevent control center overheating. A strategically positioned HAF fan mounted on the enclosure's side creates an airflow path drawing outside air through the large vent, across the electronics, and exhausting it into the greenhouse. This dual-purpose design simultaneously cools the control components while introducing fresh air into the growing environment.

7.6.3 Access Panel

A cutout from the side panel around the control center allows for the placement of a door providing easy access to the electrical components. This door will also incorporate the

louvred vent, enabling proper ventilation for the electrical components and acting as the intake for the HAF fan connected to the control center.

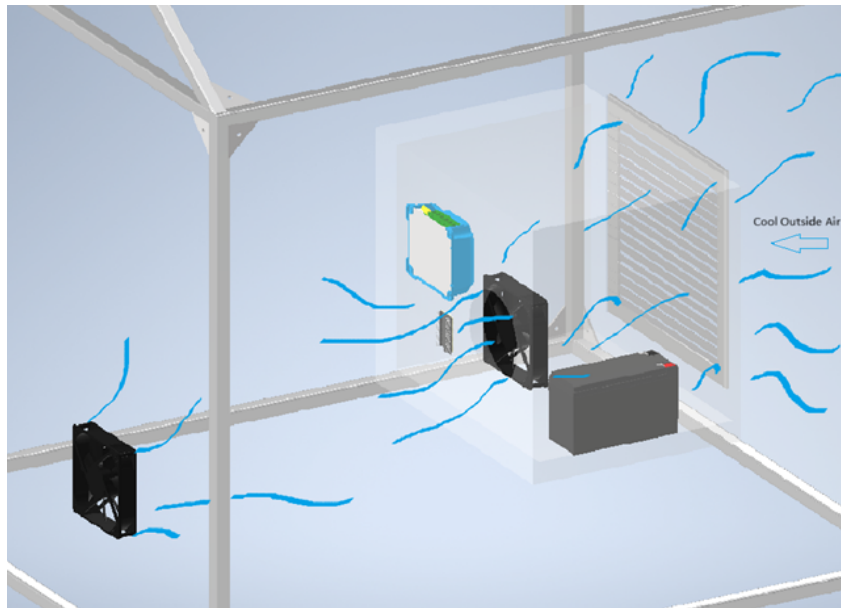


Figure 7.4: Air Intake

7.7 Environmental Monitoring

The greenhouse uses a network of environmental sensors connected to a central microcontroller to continuously monitor key parameters such as temperature, humidity, soil moisture, light intensity, and CO₂ concentration. These measurements provide real-time feedback on the internal climate conditions, which are essential for optimizing plant health and growth.

The data collected is processed by the microcontroller and used both for local decision-making (e.g., activating ventilation or irrigation systems) and for remote monitoring via IoT platforms. Through cloud-based interfaces, users can access historical data and current conditions, and adjust system settings as needed. The following sections detail the selection and implementation of the microcontroller, the individual sensors, and the data acquisition platforms used in the system.

7.7.1 Controller

The ESP32 microcontroller, developed by Espressif, is selected for its abundant I/O options, making it ideal for managing multiple sensors and actuators through its numerous GPIO pins. Its built-in Wi-Fi and Bluetooth connectivity enables seamless communication with user interfaces, allowing for real-time monitoring of the greenhouse environment.

The ESP32 supports a variety of communication protocols such as I²C, UART, and SPI, allowing seamless integration with environmental sensors and control modules. Its dual-core processing power ensures efficient real-time data handling and decision-making. Its integration with Arduino's IDE software allows for easy programming capabilities. Additionally, the ESP32's low power consumption and wide community support make it a

reliable and cost-effective choice for an embedded greenhouse controller.

7.7.2 Sensors

Multiple sensors enable accurate monitoring of both the internal greenhouse environment and the external ambient conditions. This data informs the control of greenhouse parameters and provides a baseline for comparison between internal and external climates.

The external temperature and humidity sensor will be exposed to direct sunlight and rainfall, while the internal sensors will encounter both sunlight and periodic misting. Therefore, appropriate protection is essential to ensure sensor longevity. Custom 3D-printed Stevenson screens can be employed to shield the sensors from water exposure and overheating, while still permitting adequate airflow for accurate environmental readings through the slatted design. White 3D-printed material should be used to reflect the most sunlight and prevent overheating. The Stevenson screen should be placed at least 1.2 m from the ground (?).

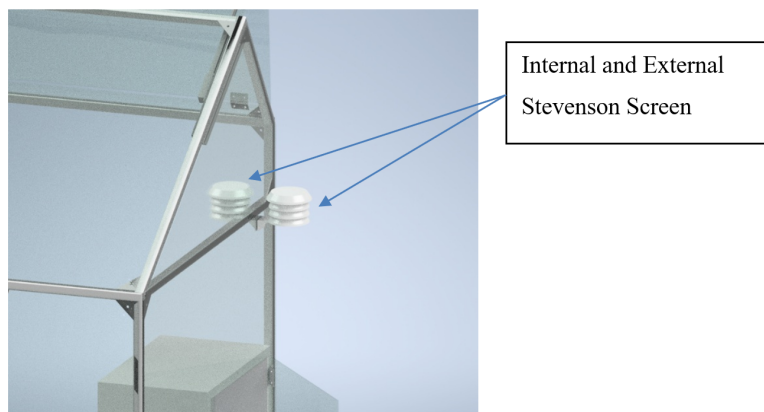


Figure 7.5: Stevenson Screens for sensor protection

The following sections break down each sensor's selection and operation.

7.7.2.1 Temperature and Humidity Sensor

The DHT22 sensor was selected for its balance of accuracy, low cost, and ease of integration. Since it measures both temperature and relative humidity, it reduces the need for separate sensors, streamlining wiring and system design. It communicates with the ESP32 using a single-wire digital interface, making implementation straightforward.

Table 7.1: DHT22 Temperature and Humidity Sensor Specifications

Parameter	Specification
Temperature Range	-40 to 80 °C
Temperature Accuracy	±0.5 °C
Humidity Range	0–100 % RH
Humidity Accuracy	±2–5 % RH
Operating Voltage	3.3–5.5 V
Output Signal	Digital (single-wire protocol)
Sampling Period	≥ 2 seconds

7.7.2.2 Light Sensor

The BH1750 light sensor is chosen for its good balance of accuracy, cost-effectiveness, and ease of integration. It provides precise ambient light measurements in lux, making it suitable for monitoring plant lighting conditions with a simple conversion to PPFD values. Its I²C communication protocol allows seamless and reliable connection with the ESP32, using only two GPIO pins. The sensor also requires minimal configuration and is supported by well-documented libraries, enabling quick setup and consistent performance in greenhouse environments.

Table 7.2: BH1750 Light Sensor Specifications

Parameter	Specification
Operating Voltage	3.3–5V
Interface	I ² C (7-bit address: 0x23 or 0x5C)
Measurement Range	1–65,535 lx
Resolution	1 lx (High-Resolution Mode)
Accuracy	±20% at 25°C
Current Consumption	≈0.12 mA (active), 0.01
Operating Temperature	–40°C to +85°C

7.7.2.3 Carbon Dioxide Sensor

The MH-Z19 CO₂ sensor is chosen due to its balance of affordability, accuracy, and ease of integration with microcontrollers like the ESP32. It uses Non-Dispersive Infrared (NDIR) technology to detect CO₂ concentrations, which provides reliable and stable readings with minimal calibration.

The sensor communicates through UART or PWM, offering flexibility in how data is read by the system. Its compact design and low power consumption (typically 33 mA at 5 V) make it ideal for continuous indoor monitoring in environments such as greenhouses. The

sensor has a measurement range of 0–5000 ppm with an accuracy of $\pm(50 \text{ ppm} + 5\% \text{ of reading})$ in the 0–2000 ppm range.

Table 7.3: MH-Z19 CO₂ Sensor Specifications

Parameter	Specification
Technology	NDIR (Non-Dispersive Infrared)
Communication	UART, PWM
Operating Voltage	3.6–5.5 V
Current Consumption	33 mA (typical)
Measurement Range	0–5000 ppm
Accuracy	$\pm(50 \text{ ppm} + 5\% \text{ of reading})$
Warm-up Time	3 minutes
Operating Temperature	0–50 °C

7.7.2.4 Soil Moisture Sensor

The soil moisture sensor provides real-time monitoring of substrate hydration levels, enabling the automated misting system to deliver precise irrigation when needed. The SEN0114 soil moisture sensor is selected for its simplicity, cost-effectiveness, and compatibility with the ESP32 microcontroller.

Table 7.4: SEN0114 Soil Moisture Sensor Specifications

Parameter	Specification
Output Signal	Analog voltage
Operating Voltage	3.3–5 V
Interface	1-wire (requires ADC)
Probe Material	Corrosion-resistant alloy
Measurement Depth	Up to 37 mm

The sensor operates by measuring the conductivity between two exposed probes inserted into the soil, where higher conductivity indicates greater moisture content. It provides an analog voltage output (0–3.3 V) corresponding to the soil's moisture level, allowing for real-time monitoring of water content in the growing medium.

The sensors ease of integration, low power requirements (drawing less than 20 mA) make it well-suited for irrigation automation. The sensor only requires one wire for communication and connects to an analog-to-digital converter (ADC) pin on the ESP32. Initial calibration is required to establish the relationship between voltage output and actual soil moisture content. This can be achieved as follows.

Calibration Procedure

1. **Hardware Setup:** Connect the soil moisture sensor's output to an ADC pin (e.g., GPIO 36) on the ESP32. Ensure the sensor is powered correctly.
2. **Wet Point Calibration:** Place the sensor probes in fully saturated soil (considered 100% moisture). Record the stable ADC reading, R_{wet} .
3. **Dry Point Calibration:** Place the sensor probes in completely dry soil (considered 0% moisture). Record the stable ADC reading, R_{dry} .
4. **Software Implementation:** In the microcontroller code, map subsequent ADC readings (R) to a percentage value using the following linear interpolation formula:

$$\text{Moisture Percentage} = \left(\frac{R - R_{\text{dry}}}{R_{\text{wet}} - R_{\text{dry}}} \right) \times 100\% \quad (7.4)$$

7.7.2.5 Rain Sensor

A rain sensor is mounted on the exterior of the greenhouse to detect rainfall and trigger protective actions. When rainfall is detected, the system is programmed to automatically close the ventilation openings, thereby safeguarding electrical components and preventing overwatering of the plants.

The selected model is an HKD general-purpose rain sensor, which features a large sensing surface to improve droplet detection. The sensor operates by using exposed conductive tracks on the board: when water bridges the tracks, the resistance changes significantly—from approximately 2 M Ω when dry to around 100 k Ω when wet. This variation is processed by the module, which outputs a digital signal (logic HIGH or LOW) to indicate the presence of rain.

7.7.3 Data Acquisition and Monitoring

The ESP32 can collect sensor data and store it locally in .xls format on an SD card. However, to provide a more accessible and user-friendly interface, data will instead be uploaded to a cloud-based platform such as ThingSpeak. ThingSpeak supports monitoring and control of up to 8 fields with a minimum update interval of 15 seconds. The free tier offers up to 3 million messages per year, which equates to approximately 8,000 messages per day. Alternatively, Adafruit IO provides similar cloud-based data logging capabilities along with customizable dashboards. Its free tier includes 30 data points per minute, 30 days of data retention, action triggers every 15 minutes, and a limit of 5 dashboards. Blynk is another free IoT platform that allows for easy creation of mobile apps to control and monitor hardware remotely. It is fully compatible with the Arduino platform, making it a convenient choice for IoT projects.

Local Monitoring

Local monitoring is useful when a user is on-site near the greenhouse and wants to easily view the current internal conditions. Additionally, in the event of a Wi-Fi outage, the greenhouse can still communicate essential data locally. For this reason, a small LCD

screen is used to display key environmental parameters, including temperature, humidity, light levels, soil moisture, and CO₂ concentration. The LCM1602A display is selected for its built-in I²C communication interface. This allows it to share the same two pins already used by the BH1750 sensor, since I²C supports multiple devices on a single bus through unique addresses. The LCM1602A operates at address 0x3E, enabling efficient integration while reducing GPIO pin usage compared to the standard 8-bit parallel interface. The screen cycles through these readings to present the information clearly and efficiently. It is mounted on the control room door and is visible through the Perspex panel, allowing for quick visual access without needing to enter the enclosure.



Figure 7.6: LCD screen for local monitoring

7.8 Power Supply

The power supply provides energy to the system as a whole and is crucial for the general operation and reliable functioning of the device. Power is supplied by a 12 V battery through the integration of a barrel jack. However, several components require 5 V or 3.3 V for operation. Therefore, voltage conversion is necessary to supply the correct operating voltages. Instead of a traditional linear voltage regulator, a buck converter is chosen for its higher efficiency, especially when stepping down from 12 V. Two fuses are incorporated to safeguard the components against potential failures or voltage spikes. Following the guideline of using 1.5 times the maximum expected current, a 15 A fuse is installed to protect the 12 V line supplying high-power components such as the solenoid valve and heater. Additionally, a smaller 5 A fuse is placed on the supply line to the buck converter, ensuring protection for the lower-power electronics operating at 5 V and 3.3 V.

7.8.1 Step-down Buck Converter

A buck converter is a type of DC-DC converter that efficiently reduces a higher input voltage to a lower output voltage using high-speed switching elements (typically a transistor and a diode) along with energy storage components like inductors and capacitors. This method significantly reduces power loss compared to linear regulators, which dissipate excess voltage as heat.

For this application, an MP4560 buck converter module is used. It is a high-efficiency, step-

down switching regulator capable of handling input voltages up to 36V and outputting up to 3A. It is configured to provide a steady 5V output, which is distributed via a 5V rail for convenient access by various components. The MP4560 features an integrated high-side switch, built-in thermal protection, and short-circuit protection, making it well-suited for embedded applications.

An LED is included in the 5V power circuit to indicate active power supply, giving the user a visual cue that the system is powered. In accordance with the MP4560's datasheet recommendations, decoupling capacitors are placed on both the input and output to stabilize voltage and suppress noise. These capacitors serve to filter out voltage ripples and transient fluctuations by storing charge and releasing it when needed, ensuring clean and stable voltage to connected components.

7.8.2 Battery Capacity

A battery capacity of at least 70Ah is required to power the average energy usage of the system and its components as shown in the calculations in Appendix D . Thus to provide for DoD depth of discharge allowance minimum discharge of 20% for safe lithium battery usage a battery capacity of 100Ah is selected.

7.8.3 Solar System

A solar system rated to provide at least 240W would be able to charge the battery fully in one day assuming a moderate system efficiency of 80% as shown in the calculations. Thus a single solar panel of at least 240W or two 120W panels can be sourced.

7.8.4 Provided Equipment

The M&M department of engineering has provided a 12V, 100Ah battery which should be sufficient based on the previous DoD calculations. Additionally two solar panels rated at 105W are provided along with a Skyking Solar Charge Controller. This charge controller will be able to charge the battery from the 2 solar panels which will power all the components.

7.9 Relays

The ESP32 operates with 3.3V digital logic, which is suitable for interfacing with most sensors. However, it cannot directly control components that require higher voltages or draw significant current. To address this, relay modules are used to enable the ESP32 to switch high-power devices using low-voltage control signals. Two 4-channel relay module has been selected for this project, capable of switching loads up to 250V and 10A, and compatible with 3.3V logic levels. This module will allow the ESP32 to control 12V devices such as fans, the linear actuator, and the heater efficiently and safely.

7.10 Linear actuator and H-Bridge

To actuate the vent, a linear actuator is required. The Actuonix L16 S linear actuator is selected for this purpose due to its 12V operating voltage, its availability, and ability to deliver up to 50N of force, more than sufficient to lift the panel. The actuator includes built-in limit switches that stop motion at full extension or retraction. To control both extension and retraction, the polarity of the 12V power supply must be reversed. For this reason, an H-bridge circuit is necessary. Instead of using a dedicated H-bridge driver such as the DRV8871, the design implements two switching relays wired in an H-bridge configuration. By toggling the relays, the polarity of the 12V supply to the actuator can be reversed, enabling both extension and retraction. Careful coding is essential to ensure that both relays are never activated simultaneously, as this could create a short circuit across the supply. Proper logic sequencing in the microcontroller program guarantees safe operation while maintaining reliable control of the actuator. 3D-printed components can be used to mount the linear actuator to the Perspex panels, providing two degrees of freedom to accommodate slight pivoting during extension and retraction.

8 System Integration

This section of the report outlines the plan for integrating both the physical hardware components and the software architecture of the greenhouse system. It begins with the wiring layout of all electronic components, followed by a description of the software framework and control logic that will govern the system operation. This section on system integration demonstrates how the various subsystems, sensors, actuators, and controllers, work together to achieve the overall functionality of the greenhouse. It also provides a clear roadmap for implementation, troubleshooting, and future scalability.

8.1 Wiring

The battery provides a 12V power rail to any components that require it. This 12V supply is connected to the Common (COM) terminals of the relay module. When activated by the microcontroller, each relay allows current to flow from the COM terminal to the Normally Open (NO) terminal, supplying 12V to the connected device. On the input side of the relay module, the VDD pin must be connected to a 5V power source, and the GND pin to ground. The four input pins labeled IN1–IN4 are connected to general-purpose I/O (GPIO) pins on the microcontroller, specifically GPIO 38,37,47 and 48, allowing the relays to be toggled via 3.3V logic signals.

These relays control the 12V-powered components: airflow fans, heater fan, grow light, and solenoid valve. Each of these devices requires a connection to ground on their negative terminal, which is wired to the negative terminal of the battery. A common ground rail ensures all components share the same electrical ground reference. The wiring schematic can be seen in the figure below

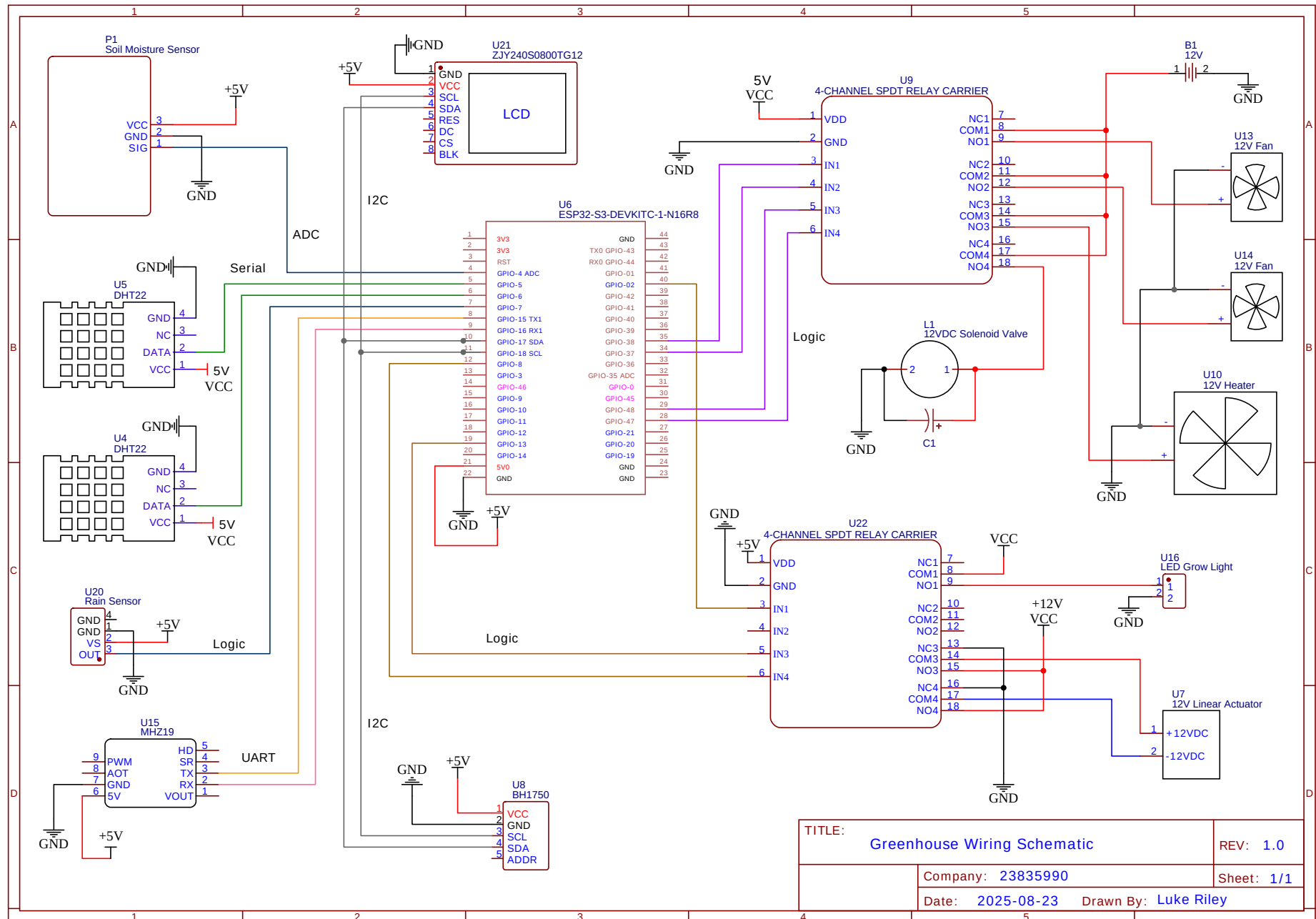


Figure 8.1: Wiring schematic

For the linear actuator, bidirectional motion is achieved by reversing the polarity of the supply voltage using two channels of the relay module. GPIO 8 and GPIO 13 of the microcontroller are connected to input pins IN3 and IN4 of the relay module, respectively. The 12V supply is wired to the Normally Open (NO) terminals of both relays, while the actuator leads are connected to the Common (COM) ports. The Ground connection is tied to the Normally Closed (NC) terminals of each relay. With this configuration, switching the relays in different states alternates the polarity applied to the actuator, enabling both extension and retraction.

Other components and sensors require a 5V supply. A buck regulator is used to step down the 12V input efficiently, avoiding the high energy loss associated with voltage dividers. The regulator's VIN is connected to the 12V rail, and its GND pins are tied to ground. The VOUT pin is connected to a 5V power rail that supplies all 5V-dependent components. Before connecting any loads, the regulator should be calibrated by measuring the output with a multimeter and adjusting the onboard potentiometer until the desired 5V output is reached.

The DHT22 sensors each require a 5V VCC connection, ground to GND, and a data line connected to GPIO 5 and 6 respectively, using digital communication. The soil moisture sensor is similarly powered with 5V to VCC, ground to GND, and its data line connected to GPIO 4. The BH1715 light sensor communicates via the I²C protocol and is connected to 5V and ground, with SDA and SCL lines wired to GPIO 17 and 18 respectively. The rain sensor outputs a logical high voltage level of 3.3V when water is detected, accordingly GPIO 7 is set as an input pin to read the logic level. The CO₂ sensor requires a 5V supply to its 5V pin, GND to ground, and TX/RX lines connected to UART-compatible GPIO pins—GPIO 15 and 16 in this case.

For local display of environmental readings, an I²C LCD module is used. The module requires 5V on its VCC pin and ground on GND. Communication with the microcontroller is achieved via the I²C protocol, with GPIO 17 connected to the SDA line and GPIO 18 to the SCL line. This configuration reduces the number of required connections compared to a parallel LCD, while still enabling reliable display of real-time sensor data. A wiring schematic is provided below.

8.2 Software Framework and Control Logic

This section outlines the software design framework for the greenhouse system. The accompanying flowcharts illustrate the sequence of tasks and decision-making processes implemented on the microcontroller. These flowcharts serve to simplify the development of complex control algorithms, particularly those involving multiple variables with competing operational requirements.

In certain situations, control actions are prioritized based on environmental conditions. For example, if the internal temperature exceeds the setpoint while soil moisture levels remain adequate, the misting system may still be activated to assist with cooling. In such cases, temperature regulation is prioritized over soil moisture conservation.

The control logic is structured into distinct modules, arranged in approximate order of priority: *Principal Control*, *Temperature Control*, *Airflow Control*, *Irrigation Control*, *Lighting Control*, and *Air Quality Control*.

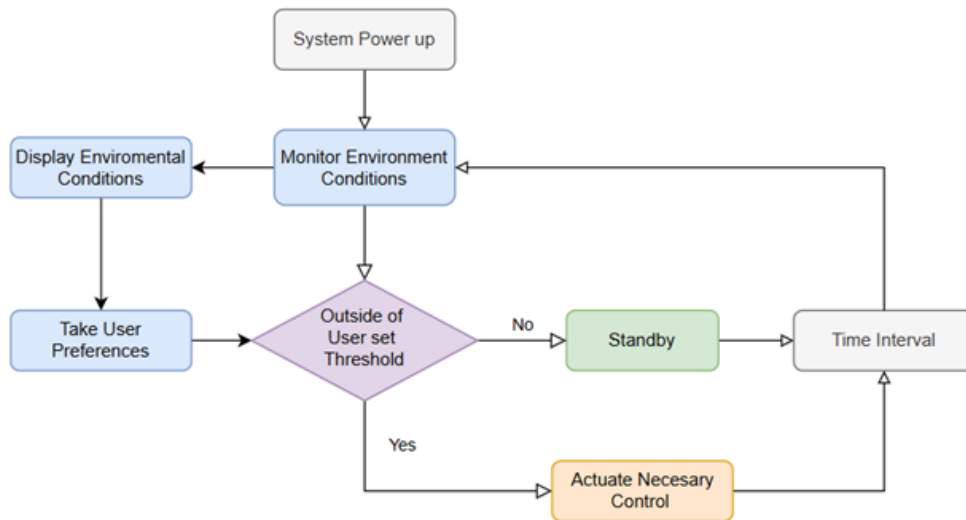


Figure 8.2: Principal Control Flowchart

Principal Control involves monitoring internal and external environmental conditions, to inform decisions about which components need to be actuated in response to undesirable conditions. These environmental readings are displayed to the user, who can then set custom threshold values based on the specific requirements of the plants being cultivated. When any measured condition falls outside the user-defined thresholds, the system responds by activating the appropriate control mechanisms. If no user preferences are provided, the system defaults to a broad, general-purpose set of control setpoints suitable for common plant types.

The inclusion of these flowchart diagrams streamlines the programming of the microcontroller's control logic by providing a clear framework for implementation, reducing the likelihood of debugging issues caused by conflicting decision-making processes.

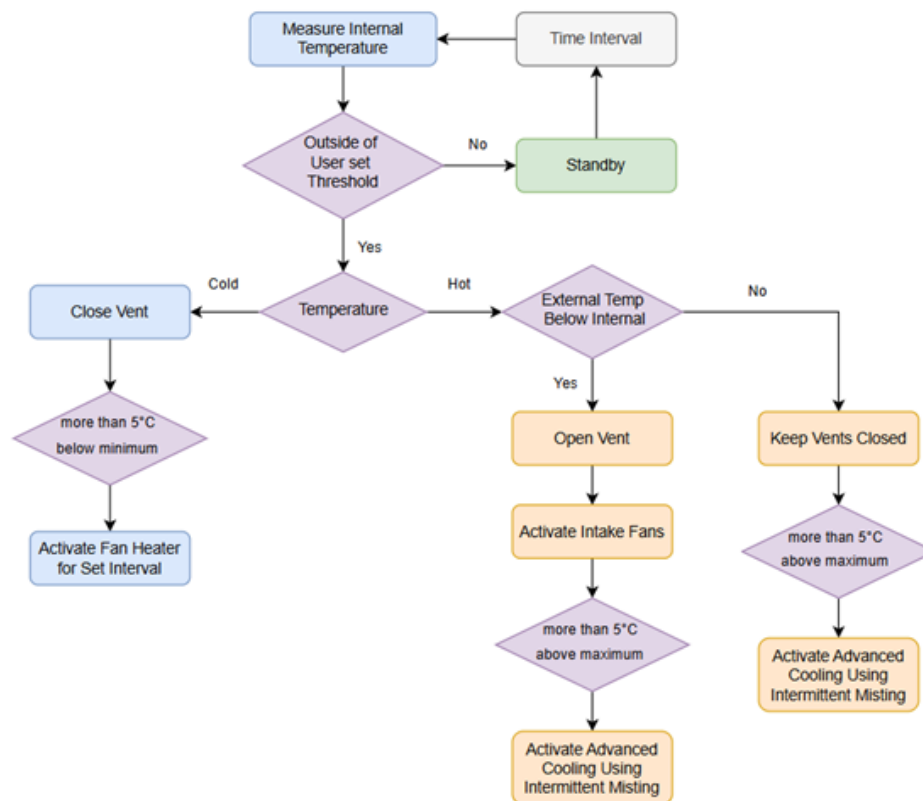


Figure 8.3: Temperature control

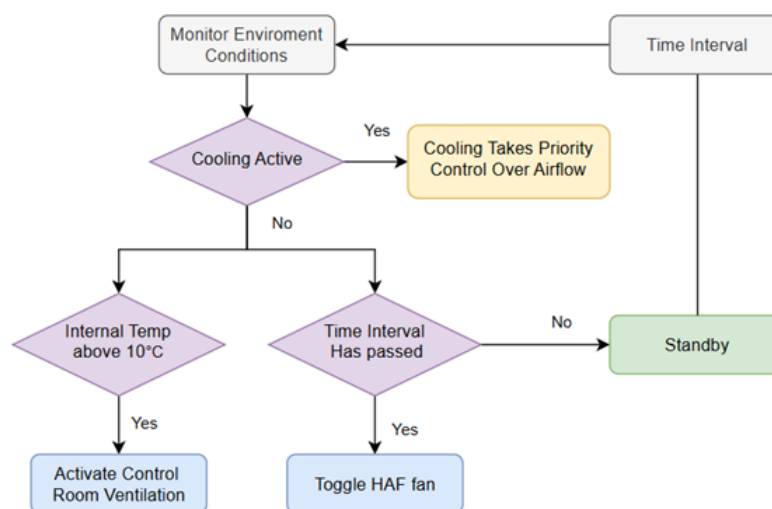


Figure 8.4: Airflow control

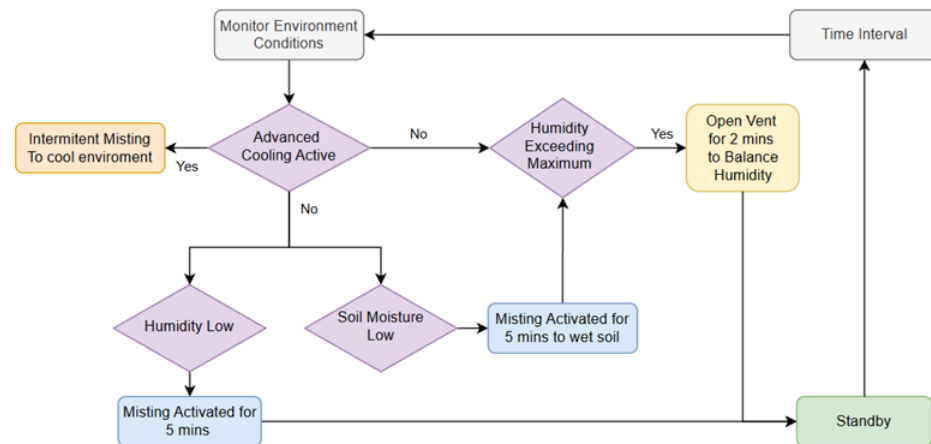


Figure 8.5: Irrigation control

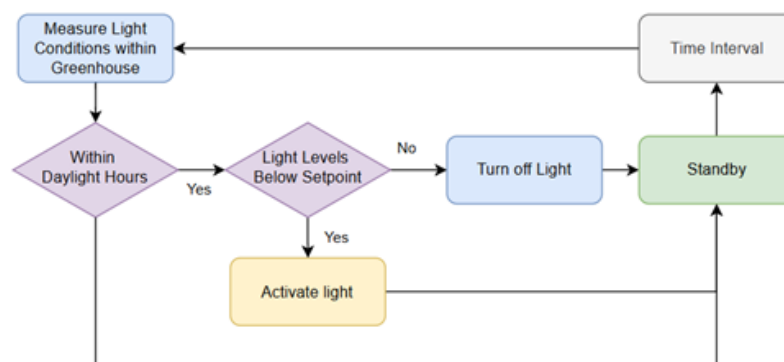


Figure 8.6: Light control

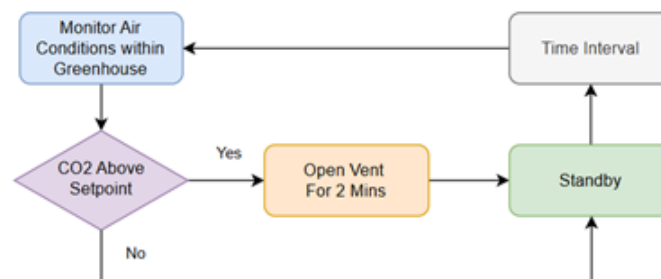


Figure 8.7: Air quality Control

9 Design Evaluation

This section outlines the tests used to validate the greenhouse systems setup and operation. It includes performance tests to validate selection and design of sensors, actuators the system as a whole and its wireless monitoring and control capabilities.

9.1 Sensor Validation

Each sensor needs to be tested to verify its response and accuracy to real shifts in environmental conditions. For each test, the sensor was connected to the ESP32 microcontroller with the appropriate pins configured for operation. A serial communication link was established to output the sensor readings at one-second intervals. These serial outputs were then imported into MATLAB, where the data was processed and plotted to generate the response graphs presented in the following subsections. This approach ensured consistent logging, real-time monitoring, and clear visualization of each sensor's performance.

9.1.1 Temperature and Humidity Sensor

Test Objective

The purpose of this test was to verify the accuracy and responsiveness of the DHT22 sensor to real environmental changes in temperature and humidity.

Methodology

The sensor was connected to the ESP32 microcontroller and live readings were recorded. A burst of warm air was directed towards the sensor to simulate a rapid change in environmental conditions. The corresponding temperature and relative humidity values were logged and plotted using MATLAB for analysis.

Results

The plotted data, shown in Figure 9.1, demonstrates the sensor's response to the applied stimulus. A clear rise in temperature was observed immediately after exposure to warm air, accompanied by a corresponding change in humidity.

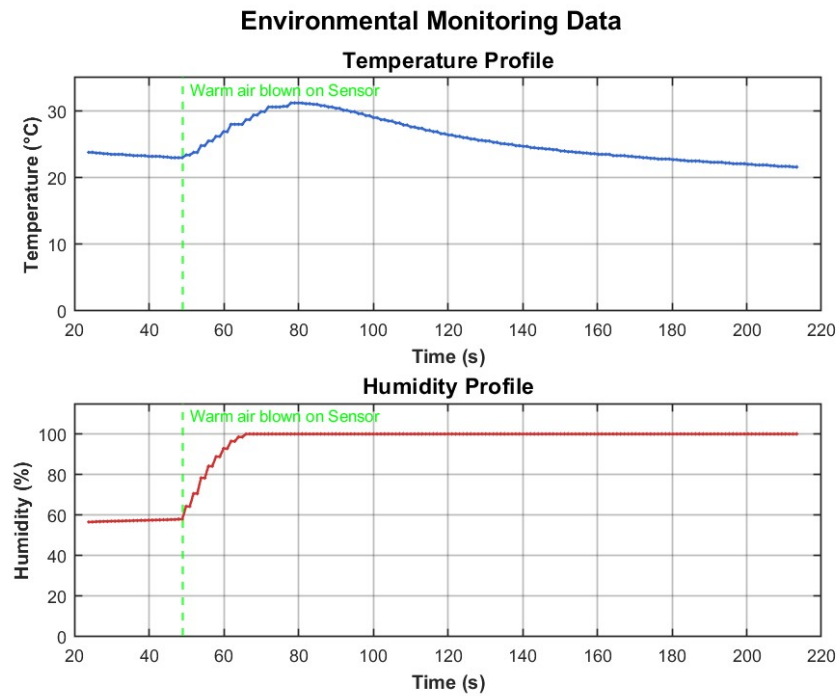


Figure 9.1: DHT22 response to applied warm air (temperature and humidity vs. time).

Discussion

The test confirms that the DHT22 sensor responds promptly to changes in environmental conditions, with smooth and continuous data output. The test showed the expected rise in both temperature and humidity. However, it was noted that the humidity readings temporarily saturated at the maximum percentage due to condensation forming within the sensor. After a short period, the values gradually returned to ambient levels. While this behavior is not ideal, it is a known characteristic of the sensor under extreme conditions and is not expected to cause issues during normal greenhouse operation, where such rapid condensation is unlikely to occur. However, slight response delays were noted, which are consistent with the DHT22's documented sampling rate and latency.

9.1.2 Soil Moisture Sensor

Test Objective

The purpose of this test was to validate the functionality and calibration response of the soil moisture sensor under different moisture conditions.

Methodology

The sensor was connected to the ESP32 microcontroller and real-time values were logged. The sensor probes were sequentially tested in three environments:

1. Open air (representing completely dry soil).
2. Damp soil.
3. Fully submerged in water.

The recorded data was plotted in MATLAB to observe the changes in moisture percentage.

Results

The results, shown in Figure 9.2, demonstrate the sensor's clear response across the three test conditions. The readings started at approximately 0% in open air, increased to around 50% in damp soil, and finally rose to about 80% when submerged in water. These values align with the expected qualitative behavior of the sensor.

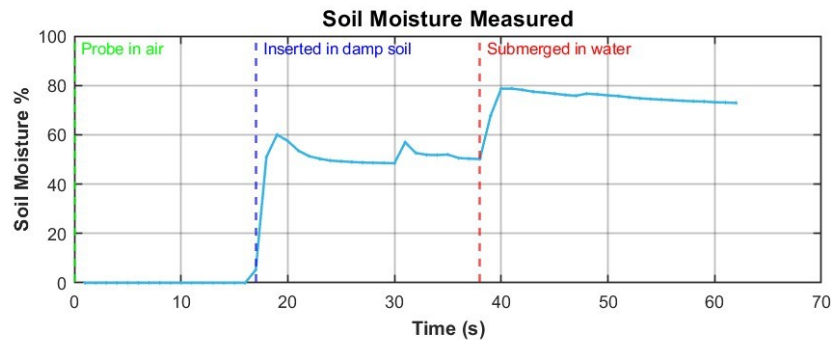


Figure 9.2: Soil moisture sensor response in open air, damp soil, and submerged in water.

Discussion

The test confirms that the soil moisture sensor is capable of distinguishing between dry, moist, and saturated conditions with clear transitions. The sensor provided consistent and repeatable readings, making it suitable for use in greenhouse irrigation control. The calibration process will ensure more accurate mapping of percentage values to volumetric water content, but the results demonstrate strong baseline performance for the intended application.

9.1.3 Light Sensor (BH1750)

Test Objective

The purpose of this test was to verify the accuracy and responsiveness of the BH1750 light sensor when subjected to sudden changes in illumination.

Methodology

The sensor was connected to the ESP32 via the I²C interface, and its lux readings were logged in real time. A torch was used as a controlled light source. The test procedure involved:

1. Measuring the baseline lux level in a dimly lit room (ambient conditions).
2. Shining a torch directly on the sensor to observe the rise in lux.
3. Removing the torch to observe the return to baseline levels.

The recorded data was processed and plotted in MATLAB to visualize the sensor response.

Results

The results, shown in Figure 9.3, indicate a sharp increase in light intensity when the torch was directed at the sensor, with the lux value rising to approximately 200. When the torch was removed, the lux level dropped back down to around 40, representing the ambient light level in the room.

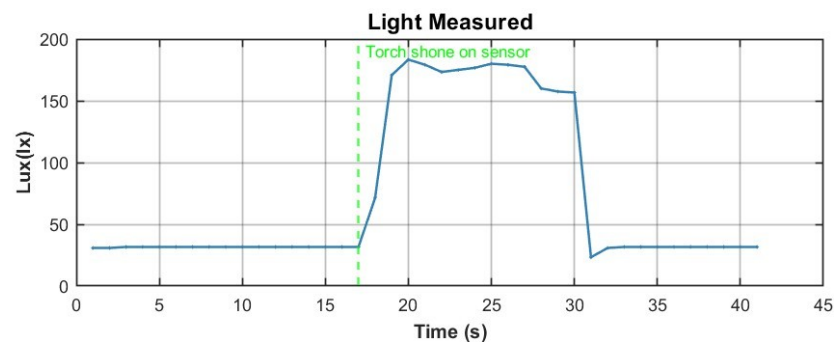


Figure 9.3: BH1750 light sensor response to torch illumination and return to ambient room lighting.

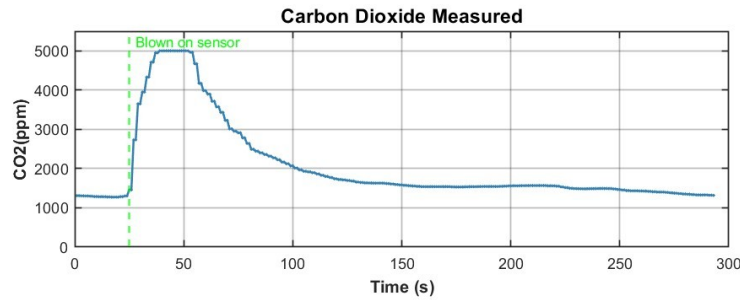
Discussion

The test demonstrates that the BH1750 sensor is highly responsive to changes in light conditions, with minimal delay between exposure and measurement. The observed values are consistent with expected lux levels, confirming that the sensor is suitable for monitoring greenhouse lighting conditions. This responsiveness ensures accurate detection of both natural and artificial light variations, which is essential for regulating plant growth environments.

9.1.4 CO₂ Sensor

To validate the CO₂ sensor, the device was first left to stabilize under ambient indoor conditions, which measured approximately 1000 ppm. A short burst of exhaled air was then directed towards the sensor to create a rapid increase in CO₂ concentration. The sensor response was logged via the ESP32 serial output at one-second intervals and plotted in MATLAB.

As shown in Figure 9.4, the concentration quickly rose and saturated at the sensor's maximum measurable value of 5000 ppm. After exposure, the readings steadily decreased and returned to baseline ambient levels of approximately 1000 ppm within less than two minutes. This confirmed both the sensor's responsiveness to rapid changes in CO₂ concentration and its ability to recover to stable baseline values after disturbance.

Figure 9.4: CO₂ sensor response

9.2 Actuator Tests

Each actuator was connected to its respective relay channel, corresponding to the assigned GPIO pin on the microcontroller. Testing was carried out by first supplying the rated voltage directly to each actuator, followed by validation through relay control. A logical HIGH signal was written to each GPIO pin to trigger the relay, with successful activation confirmed visually via the built-in indicator LED and audibly by the distinct clicking sound of the relay switch. As the relay module operates in active-high mode, actuators were connected to either the Normally Open (NO) or Normally Closed (NC) terminals as required. Once verified, the appropriate 5 V or 12 V supply was connected to the Common (CO) port to power the actuators. Table 9.1 summarizes the testing procedures and outcomes for each actuator.

Table 9.1: Actuator Testing and Observations

Actuator	Test Procedure	Observation
Solenoid Valve	12 V applied directly, then via relay control. Bulk and high-frequency capacitors added to suppress voltage spikes.	Audible switching confirmed; misting nozzles activated and deactivated with relay control.
Linear Actuator	12 V applied with polarity switching, then via relay-controlled H-bridge.	Extension and retraction observed correctly in both directions. Relay control verified reliable actuation.
Supplemental Lighting	5 V applied directly, then switched via relay control.	Purple UV grow light switched on and off reliably under both direct and relay control.
Intake / HAF Fans	12 V applied directly, then via relay control.	Fans powered on successfully; airflow observed. Relay switching confirmed proper operation.

With the mains water supply connected, the solenoid valve successfully controlled the misting system, providing adequate pressure to the misting nozzles. The nozzles were observed to start and stop spraying water in direct response to the relay switching.

During testing, it was observed that switching the misting solenoid simultaneously with Wi-Fi activity (e.g., ThingSpeak uploads) introduced supply transients that caused the ESP32 to briefly brown-out and reset. The solenoid's inductive kickback and relay contact bounce injected noise onto the 12 V rail, which coupled into the 5 V supply. To mitigate this, a bulk electrolytic capacitor and a high-frequency ceramic capacitor were added at the solenoid driver terminals. These components damped the transient response and provided local energy buffering, resulting in stable system operation without timeouts or resets.

9.3 Remote monitoring

Remote monitoring enables greenhouse conditions to be observed wirelessly, removing the need for the user to be physically present near the system. The setup was implemented using ThingSpeak, with the microcontroller configured to upload sensor data for all parameters at 20-second intervals. By accessing the ThingSpeak platform, the user can view the data remotely in real time. A dashboard was also created to present the information in a clear and accessible format, as shown in the figure below. Thingspeak also allows the storage and collection of data. this data is saved and interpreted in section 9.6.

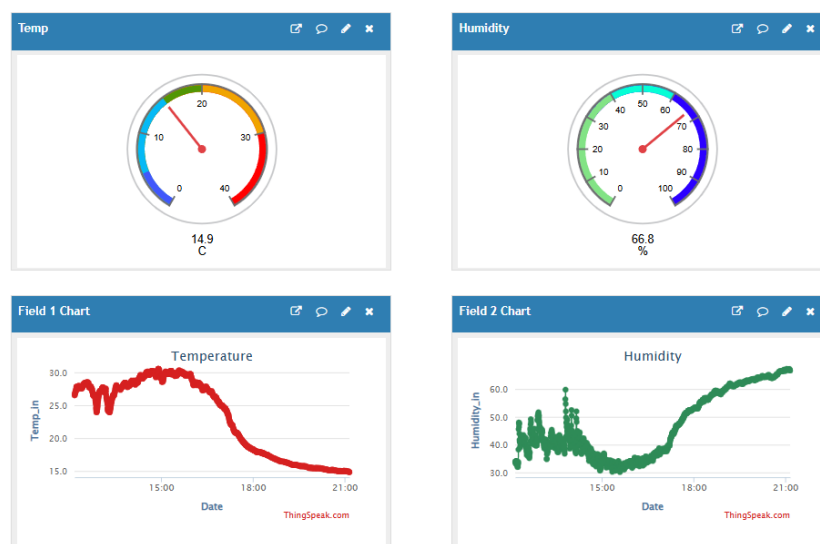


Figure 9.5: Wireless Monitoring Dashboard.

To enable user-defined environmental control, the system integrates wireless connectivity via the Blynk IoT platform. Blynk allows intuitive dashboard creation on mobile devices, giving users the ability to adjust greenhouse parameters such as temperature and humidity directly from their phones using sliders and buttons.

Although Blynk lacks the long-term data storage and analytical features of ThingSpeak, it excels in real-time control and user interaction. The ESP32 was configured with a unique Blynk template and authentication token, allowing remote updates of setpoints for various environmental parameters.

During calibration, it became clear that some user-defined values such as an excessively low humidity limit could be impractical due to natural temperature-humidity relation-

ships and fluctuations. As a result, fixed setpoints were used during testing to ensure consistent system performance. Nonetheless, the app proved valuable for user engagement and fine-tuning, offering flexibility for future customization based on plant type or user preference. Figure 9.6 shows the configured Blynk dashboard.

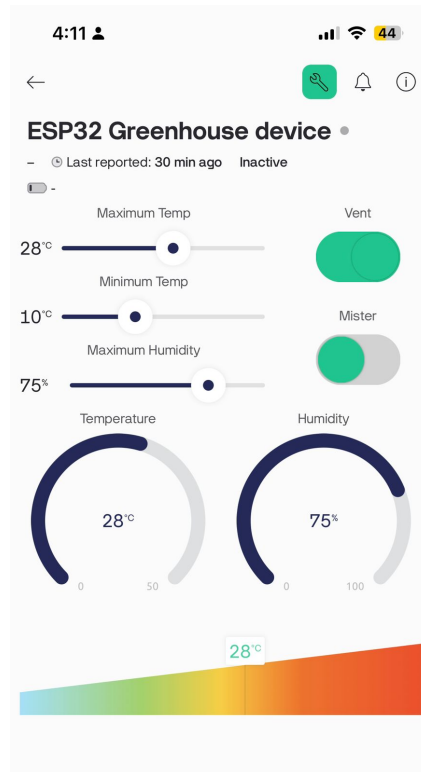


Figure 9.6: User App with Blynk

9.4 Solar Power Test

To validate the capability of the solar subsystem to power the greenhouse entirely off-grid, a series of tests were conducted. The procedure was as follows:

1. The battery voltage was measured with no load connected, showing 12.8 V, indicating a partial state of charge (full charge is approximately 14.8 V).
2. The battery was connected to the Skyking charge controller via the positive and negative terminals. Upon connection, the controller powered on and displayed the battery voltage.
3. Using the controller menu, the float voltage was set to 13.7 V, the discharge stop voltage to 10.8 V, and the battery type configured as a sealed gel battery, in accordance with the battery manufacturer's recommended operating conditions.
4. The solar panels were connected using MC4 connectors. The charge controller confirmed successful charging through its display.
5. The 12 V supply terminal block and greenhouse system ground were connected to the load terminals of the charge controller.

6. The controller display indicated power transfer to the load. This was verified by measuring the potential difference across the load terminals with a multimeter.

These steps confirmed correct wiring and successful operation of the solar power subsystem. The charge controller additionally provides low-voltage protection by disconnecting the load if the battery voltage drops below a safe threshold. During full system testing, it was confirmed that the solar array supplied sufficient power to operate the greenhouse without discharging the battery to unsafe levels.

9.5 Greenhouse Setup

The Full Greenhouse setup can be seen in the figure below. Please see Appendix [E](#) for additional pictures.



Figure 9.7: Greenhouse Setup

9.6 Enviromental Response

With the all of the sensors, actuators, solar power and monitroing sytems tested installed and integrated succesfully the system can be tested with all systems operational. the greenhouse can be powered on and allowed to measure the enviromental conditions and try control the environment to desirable conditions.

9.6.1 Dual Phase Test

To evaluate the performance of the greenhouse control system, a two-phase test procedure was conducted over a period of six days. In the first phase, the greenhouse was operated for three full day-night cycles with all sensors and monitoring systems active, but without the actuators connected. This allowed the internal environment to respond passively to

external conditions, providing a baseline measurement of natural fluctuations in temperature, humidity, and light.

In the second phase, the actuators (ventilation, irrigation, and heating) were connected and enabled. The greenhouse was again run for three full day–night cycles, with the control system actively attempting to regulate conditions towards the desirable setpoints. This phase demonstrates the effectiveness of the system in stabilizing the greenhouse environment.

The temperature, humidity and soil data from both phases were collected and plotted, enabling a direct comparison between passive response and active control. The resulting figures highlight the extent to which the actuators improve environmental stability and overall system performance.

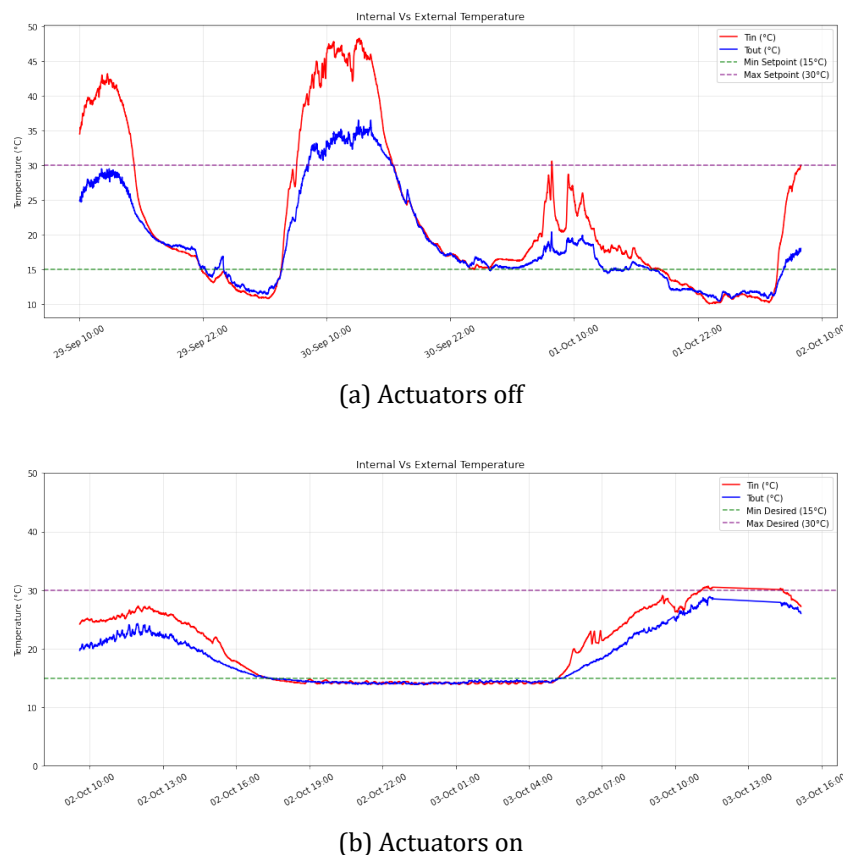
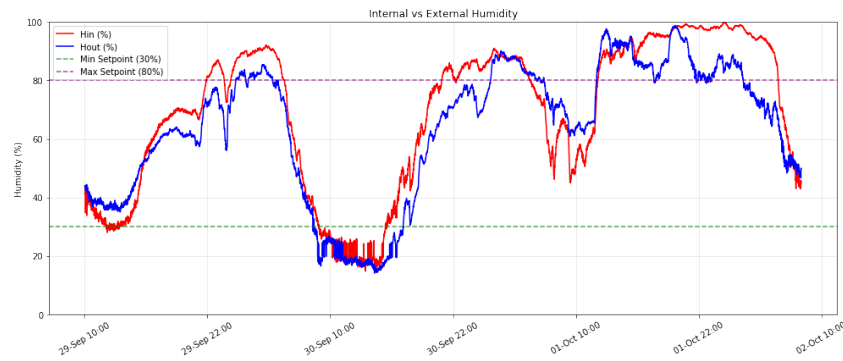


Figure 9.8: Internal vs. external temperature profiles for the greenhouse with (a) actuators off and (b) actuators on.

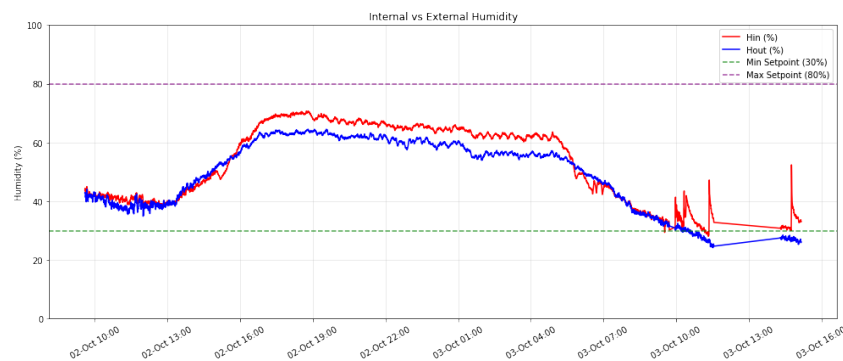
It was observed that the greenhouse system experienced difficulty in adequately cooling the internal environment during hotter daytime conditions. The internal temperature consistently rose to approximately 5 °C above the ambient external temperature. Nevertheless, the active ventilation and circulation fans provided a noticeable cooling benefit when compared to the passive test without actuators.

During the actuator-off test, the internal temperature was allowed to increase to a peak of around 48 °C, while at night it dropped well below the desired setpoint, reaching approximately 11 °C. In contrast, during the actuator-on test, the heating system effectively

maintained the internal temperature close to the desired setpoint of 15 °C, keeping the greenhouse slightly warmer than the external environment during colder conditions.



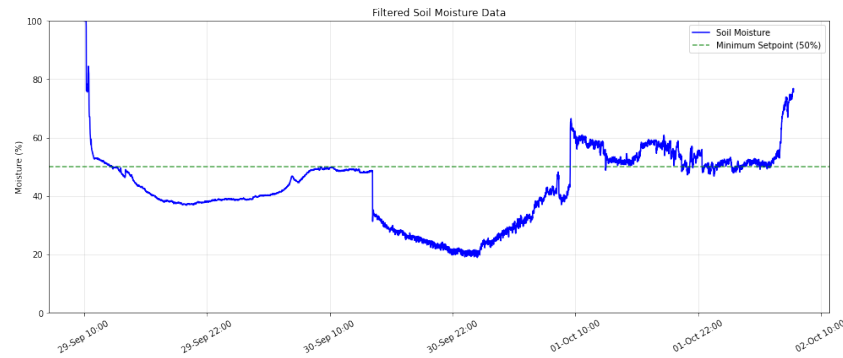
(a) Actuators off



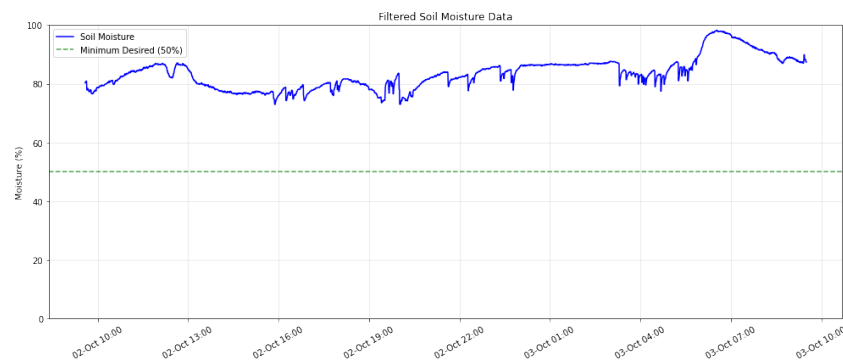
(b) Actuators on

Figure 9.9: Internal vs. external humidity profiles for the greenhouse with (a) actuators off and (b) actuators on.

The humidity exhibited a wide variation throughout the day–night cycle, which can be attributed to the dependence of relative humidity on temperature. During periods of low humidity, the system attempted to compensate by activating the misting function every 20 minutes, this can be seen through the spikes of humidity. While this temporarily increased the humidity within the greenhouse, the levels gradually declined back toward ambient conditions as the warmer air expanded and its relative humidity decreased.



(a) Actuators off



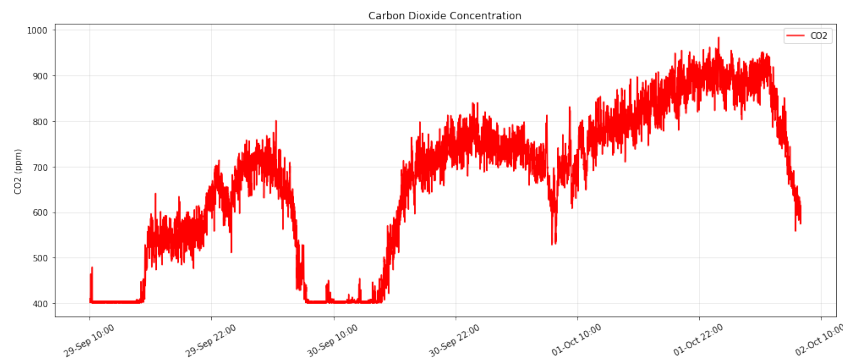
(b) Actuators on

Figure 9.10: Soil Moisture profile for the greenhouse with (a) actuators off and (b) actuators on.

During the initial phase, with the mister inactive, the soil moisture gradually dropped below the setpoint as the medium was allowed to dry out. On the third day, following a cold night, a slight increase in moisture was observed due to condensation re-entering the system.

In contrast, during the operational phase with actuators enabled, the irrigation system consistently maintained soil moisture above the desired setpoint. Levels remained stable at around 80%, never dropping below the target, demonstrating effective control and regulation of ideal moisture conditions.

9.6.2 CO₂ and Lux Observation

Figure 9.11: CO₂ Profile

The CO₂ plot provides valuable insight by highlighting a distinct day–night cycle, with concentrations rising during the night as a result of plant respiration. This trend is consistent with general ambient CO₂ behavior and does not indicate any problematic accumulation within the greenhouse. Since the levels remained within healthy ranges for plant growth, no active control or actuation was required, such as venting to regulate concentrations.

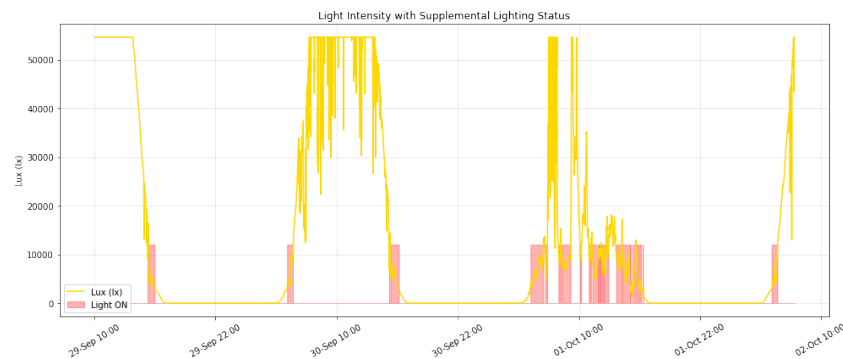


Figure 9.12: Lux Profile

The plot of the light intensity profile illustrates the digital on/off state of the supplemental LED lighting. As calculated in Appendix D, the LED was designed to provide $200 \mu\text{mol}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$ PPFD and was configured to activate whenever the ambient levels dropped below this threshold, but only during daylight hours. This corresponds to an illuminance of approximately 10 800 lux. The lighting can be observed switching on during the early morning and evening periods, as well as intermittently on the second day when cloud cover reduced incoming solar radiation.

9.7 Full system Test with Plants Growth

The operational success of the greenhouse system will still need to be validated through a controlled growth trial of microgreens, serving as a key performance indicator. seen in the figure below the test will be conducted with 4 varrieties of microgreesns; seen left to right peas, radish, beetroot and arugala microgreens have been planted.

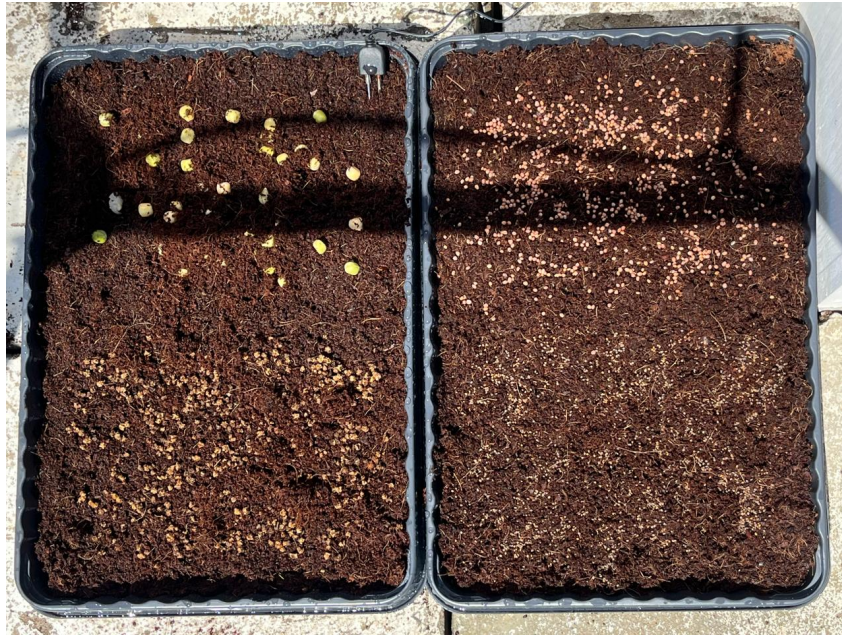


Figure 9.13: Microgreens

10 Conclusions

Future Recommendations

The results indicated that the design was effective in providing reliable monitoring of greenhouse conditions, as well as control of water supply, humidity, and supplemental lighting. However, the system showed limitations in its ability to regulate higher temperatures. For future development of an automated greenhouse in the warmer climate of South Africa, alternative cooling strategies could be explored, such as motorized shading or the integration of an evaporative cooling pad. In addition, the sizing of intake and exhaust fans could be optimized with airflow capacity in mind, favoring an overcompensated design to ensure adequate ventilation under peak heat conditions.

Conclusions

The design proved effective in monitoring all key parameters required to sustain healthy plant growth. Automated heating successfully maintained temperatures above the set-point during cooler nights, while the misting system provided sufficient control of both humidity and water supply. However, the available data was not sufficient to fully determine the system's effectiveness in preventing overheating during warmer periods.

The vent and intake fans contributed to improved airflow, helping to limit temperature rise compared to conditions without these actuators. Although overheating remained a challenge, the greenhouse system demonstrated the ability to regulate a microclimate capable of supporting plant life, while providing the user with real-time and historical data through the wireless dashboard.

In addition, the solar power system enabled fully off-grid operation, allowing the greenhouse to run entirely on renewable energy. This not only reduced the system's carbon footprint but also highlighted its potential as a sustainable solution to address food insecurity by providing controlled growing conditions, with the primary limitation being high-temperature management.

Remaining Work

The full system test with microgreens must still be carried out to confirm that the greenhouse can provide suitable living conditions for plants over an extended growth cycle. Observing plant growth responses will also provide a practical measure of system performance, complementing the sensor and actuator tests already conducted. The results of this trial will offer important insight into the system's overall effectiveness in sustaining plant health and productivity.

A Techno-economic Analysis

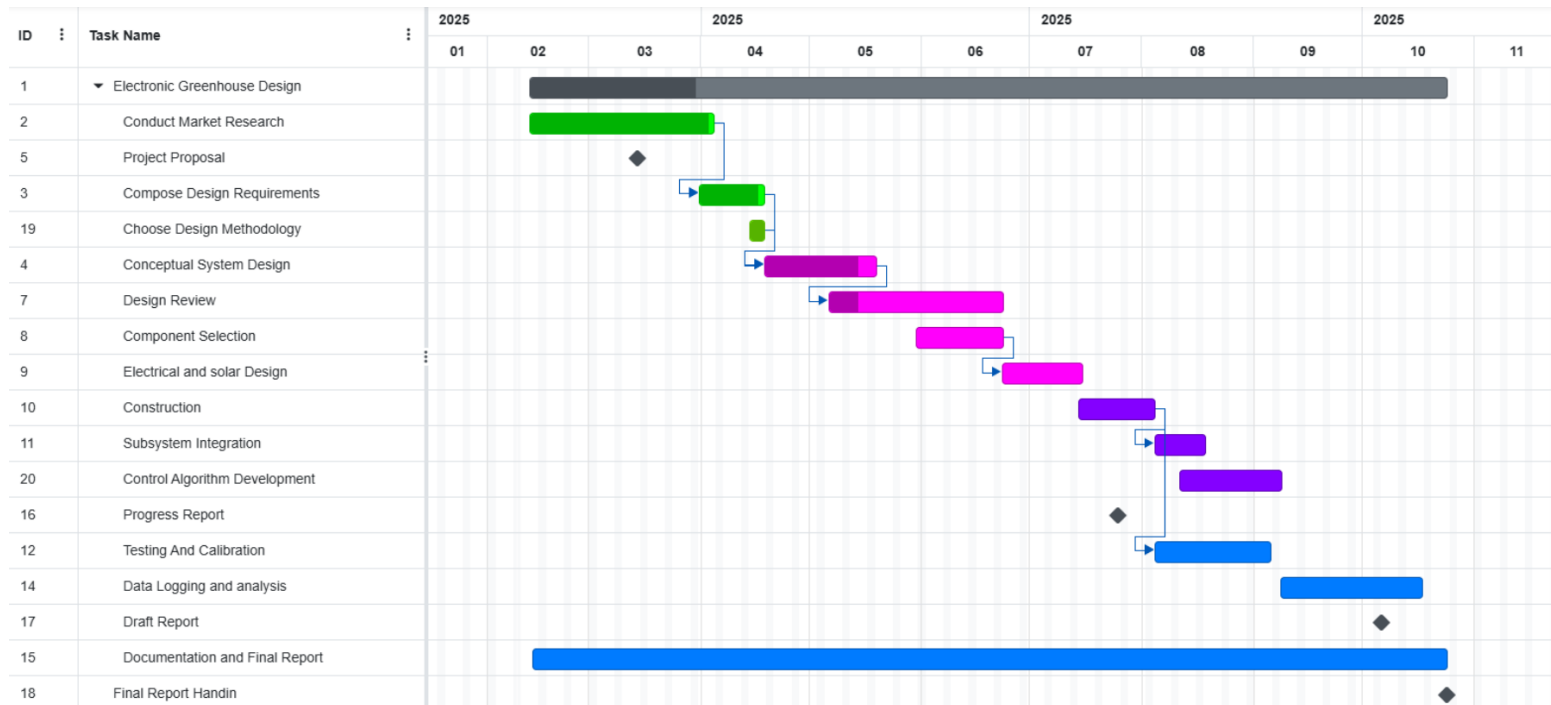
This appendix aims to discuss the project in terms of its technical and economic impact, various factors such as planning procedures, budget and ROI shall be presented.

A.1 Budget

Activity	Engineering Time		Running Costs	Facility Use	Capital Costs	MMW			TOTAL
	hr	R				Labour		Material	
	hr	R	R	R	R	hr	R	R	R
Conduct Research	65	29250	500						29750
Composing design requirements	30	13500							13500
Choose a design methodology	10	4500							4500
Conceptual system design	80	36000							36000
Design review	70	31500							31500
Component selection	25	11250			3000				14250
Electrical and solar design	25	11250			2000	10	3000	2000	18250
Control algorithm development	30	13500	200						13700
Construction	30	13500	500	500		25	7500	6000	28000
Subsystem integration	20	9000							9000
Testing and calibration	35	15750	500	500					16750
Data logging and analysis	40	18000	500	500					19000
Documentation and Final Report	60	27000	100						27100
TOTAL	520	234000	2300	1500	5000	35	10500	8000	261300

Figure A.1: Estimated Costs

A.2 Planning and Time Management



While the timeline assisted in staying on target, the main differences between planned time and effective time spent on tasks came about once the construction phase began. The physical building of the greenhouse, involving silicone sealing of all panels and construction of the wooden control room and sourcing and fastening all the right screws and bolts to ensure safe and durable operation of the actuators, ended up taking more time than anticipated. This led to a slight backlog going into the control development and calibration. However by early October this backlog was caught up and the project was back on track to have a few weeks of data logging and analysis. This was required as many adjustments based on feedback from the system were needed and fine tuning of control.

A.3 Technical Impact

The project demonstrates the feasibility of a cost-effective, automated greenhouse system that minimizes manual labour and maintenance through intelligent control and wireless monitoring. The design integrates sensors, actuators, and a microcontroller-based control system to automatically regulate key environmental variables such as temperature and humidity.

Experimental testing validated the performance of the Perspex greenhouse equipped with actuated ventilation and misting systems, both contributing to stable internal climate conditions. The system operates off-grid through a solar power supply, reinforcing its sustainability and suitability for remote or resource-limited areas.

Overall, the project establishes a foundation for scalable, low-maintenance greenhouse automation, supporting sustainable food production for users with limited time, space, or access to conventional infrastructure.

A.4 Return on Investment

The return on investment (ROI) for the automated greenhouse system can be realized through the reduced operating costs and increased crop yield stability. By utilizing a solar-powered energy system, the design offsets the electricity expenses typically associated with environmental control systems. The greenhouse automation minimizes manual labour requirements, lowering maintenance and supervision costs over time. Although the initial setup cost is higher due to the inclusion of sensors, actuators, and control electronics, these expenses are recovered through long-term operational savings and improved production efficiency. The integration of renewable energy and intelligent control thus ensures a sustainable payback period.

A.5 Potential for Commercialization

Although the small size of the greenhouse limits its use for large commercial farming, the control methods and automation developed in this project could easily be scaled for larger greenhouses. The design uses standard and easily available parts, making it simple and cost-effective to reproduce. This gives it good potential for commercialization as a small-scale product aimed at communities, schools, or home growers. The use of solar power

and automated control also makes it appealing as an educational and sustainable solution that promotes awareness of smart farming technologies.

B Safety and Mitigation Procedures

This section outlines and summarizes the safety report in accordance with the laboratory safety considerations and potential risks applicable during the construction, testing, and operation phases of the project .

B.1 Safety

The project involves the development, assembly, and testing of a small-scale electronic greenhouse system that monitors and controls environmental parameters such as humidity, temperature, light, and water supply. The system makes use of rooftop infrastructure on the M&M engineering departments solar rooftop laboratory including mains water, solar radiation, and at times grid electricity for component power supply.

B.1.1 Type of Work Involved

- Electrical wiring (5V and 12V rails)
- Sensor installation (temperature, humidity, soil moisture)
- Microcontroller programming and deployment
- Use of water lines for irrigation systems
- Mechanical assembly of the greenhouse frame and Perspex panels

B.1.2 General Lab and Rooftop Safety

1. Work within the lab may only be conducted during allocated hours.
2. Closed shoes must always be worn.
3. No loose clothing may be worn.
4. Personal protective equipment (PPE) should be worn.
5. Good housekeeping practices must be maintained during construction, testing, and operation.
6. Extreme caution should be taken when working on the sunroof, as the lab houses other projects containing sensitive components.
7. Safety harnesses must be used when working near roof edges or elevated positions.
8. The safety report must be visible and accessible while working in the lab.
9. During operation, ensure that spectators remain within designated safety areas.

10. Tools and the greenhouse structure must not block roof drains, vents, or access pathways.
11. Maintain a clear 1-meter perimeter around other ongoing rooftop activities.

B.1.3 Electrical Safety

- All wiring must be insulated, with connections clearly labeled and weather-protected.
- Use fused circuits and DC buck converters for voltage regulation (e.g., 12V to 5V).
- Only low-voltage systems ($\leq 24\text{V DC}$) will be operated.
- Grid electricity access points must be clearly labeled, and all plugs and sockets must be waterproof.
- An emergency stop circuit to disconnect all power sources is mandatory.
- Electrical systems must be powered off before performing any maintenance or component adjustments.
- Cables leading to and from the greenhouse must not create tripping hazards.

B.1.4 Water Usage and Safety

- Avoid activating the irrigation system when electrical components are exposed.
- All water connections must use leak-proof tubing and connectors.
- Ensure proper drainage on the roof to avoid slip hazards or standing water near equipment.

B.1.5 Housekeeping and Shutdown

- Remove and return all tools and hardware at the end of each session.
- Sweep up debris, zip-tie cables, and label wiring.
- Disconnect all power and water connections.
- Report damaged components or safety hazards immediately.

B.1.6 Working at Heights

The working-at-heights safety training was completed on the 2nd of April 2025 by the student. To ensure the safety of all persons involved in the project, proper working-at-heights precautions must always be observed.

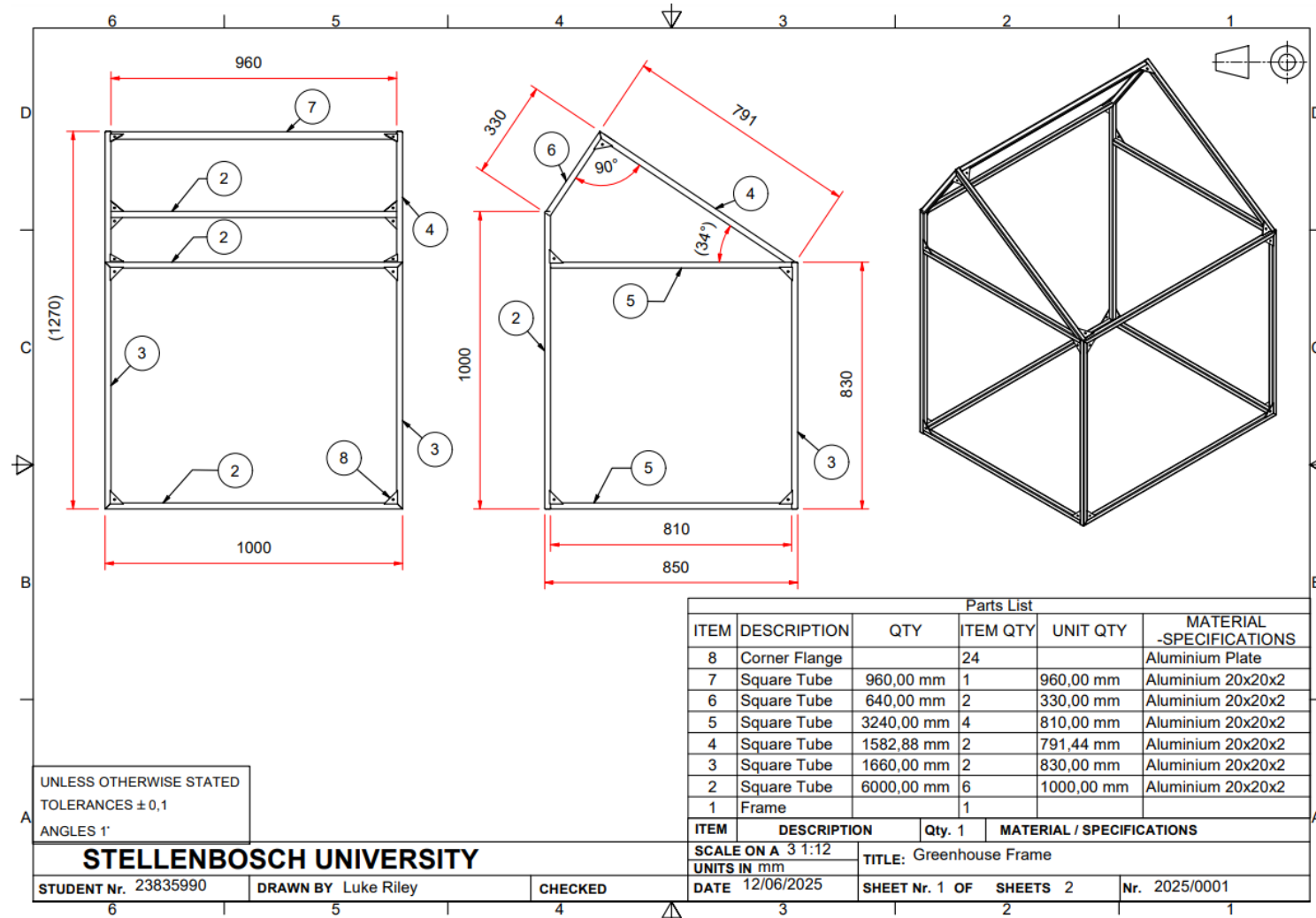
B.2 Risk Analysis and Mitigation Procedures

Table B.1: Activity-Based Risk Assessment

Activity	Risk	Risk Level	Risk Type	Mitigating Steps
Navigating lab	Tripping on loose cable or equipment	Low	P/E	Don't leave cables lying around
Work taking place on roof	Falling	Low	P	Use harness, avoid edges, complete heights training
Turning on/off equipment	Electric shock	Low	P	Check cables to ensure that insulation is intact, include warning signage
Use of hot glue gun	Accidentally contacting hot glue causing burn to skin	Medium	P	Wear protective gloves, ensure glue has enough time to cool before handling
Wiring of ports	Incorrect wiring may cause damage to components	Low	E	Make use of proper wiring diagram, label wires physically as laid out in diagram
Soldering connections of wires	Accidentally contact with the soldering iron can result in burns to the user or damage components	Medium	P/E	Keep a clean soldering workstation and use a solder stand. Turn off iron when not in use
Handling fluids near the system and operation of irrigation system	Spilling fluids can cause damage to electrical components	High	E	Do not bring fluids close to the system. Ensure irrigation system is sealed
Wiring of components	Electrical shock due to bad isolation of wires	Medium	P/E	Check all wires are well insulated before supplying power and ensure emergency stop is accessible. Include warning signage
Vent actuation	Hands in the way of moving parts causing potential small cut, bruise or pinch	Low	P	Keep hands clear of vent when operational
HAF/intake fan operation	Hands in the way of moving parts causing potential small cut, bruise or pinch	Low	P	Keep hands clear of fans when operational. Install safety mesh and coverings to prevent contact

C Engineering Drawing

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D Calculations

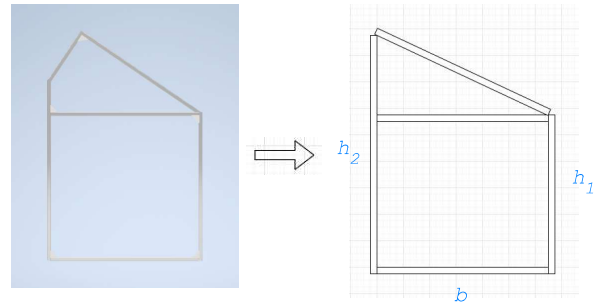
Sample Calculations:

1. Heat loss:

$h_1 := 820 \text{ mm}$ $h_2 := 1000 \text{ mm}$ height of back panel and front panel
 $l := 1000 \text{ mm}$ $b := 850 \text{ mm}$ length and base of greenhouse foundation
 $t := 0,003 \text{ m}$ thickness of panels

The shape of the side profile of the greenhouse has been approximated to simplify the calculations.

Assume ground surface area to not culminate any heat loss, but rather supply thermal mass to stabilize heat differences.



$$A := l \cdot h_1 \cdot 2 + l \cdot h_2 + \left(2 \cdot b \cdot \frac{(h_2 - h_1)}{2} + 2 \cdot b \cdot h_1 \right) \quad \text{area of all sides of perspex}$$

$$A = 4,187 \text{ m}^2$$

$$U := 5,7 \frac{\text{W}}{\text{m}^2 \text{ K}} \quad \text{for 3mm thick perspex from Maizey plastics datasheet}$$

$$\Delta T := 5 \text{ K} \quad \text{for a temperature difference of } 5^\circ\text{C}$$

$$Q := A \cdot U \cdot \Delta T \quad \text{Heat Loss} \quad (1.1)$$

$$Q = 119,3295 \text{ W}$$

$$\text{Daily} := Q \cdot 4 \text{ hr} \quad \text{asuming 4 hours of peak cold temperature differnece of } 5^\circ\text{C during the night}$$

$$\text{Daily} = 0,4773 \text{ kW hr}$$

$$\Delta T_{\text{coldest}} := (15 - 0) \text{ K} \quad \text{asumming outside temperatures can reach } 0^\circ\text{C at night and minimum setpoint temperature is } 15^\circ\text{C}$$

$$Q_{\text{coldest}} := A \cdot U \cdot \Delta T_{\text{coldest}} \quad (1.2)$$

$$Q_{\text{coldest}} = 357,9885 \text{ W}$$

2. Vent Actuation:

$$\rho_{perspex} := 1,19 \frac{\text{g}}{\text{cm}^3} \quad \text{density of perspex}$$

$$w_{vent} := 270 \text{ mm} \quad \text{width of vent}$$

$$m_{vent} := l \cdot w_{vent} \cdot t \cdot \rho_{perspex} = 0,9639 \text{ kg} \quad \text{mass of vent}$$

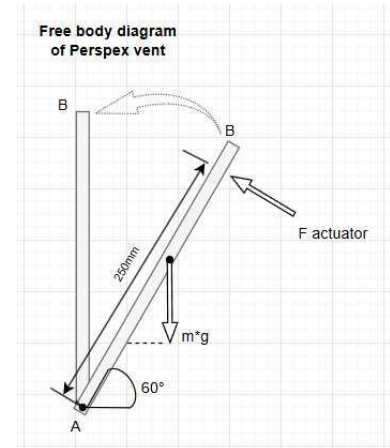
$$\theta := 60 \quad \text{angle of inclination of vent}$$

$$F_{weight} := m_{vent} \cdot 9,81 \frac{\text{m}}{\text{s}^2} = 9,4559 \text{ N} \quad \text{weight of vent acting at centroid}$$

$$\sum M_A := 0$$

$$M_{weight} := \left(-F_{weight} \right) \cdot \frac{w_{vent}}{2} \cdot \cos(\theta) = 1,2158 \text{ N m} \quad \text{moment around point A due to weight}$$

$$F_{actuator} := \frac{M_{weight}}{0,25 \text{ m}} = 4,8632 \text{ N} \quad \text{assuming that the linear actuator is connected to the panel at 250mm from the hinges at A and that it is initially acting perpendicularly to the panel}$$



3. Cooling Power:

$$\rho := 1,2 \frac{\text{kg}}{\text{m}^3} \quad \text{Air density} \quad \text{a 50 CFM small intake fan approximates to } 0.02359 \text{ m}^3/\text{s}$$

$$C := 0,6 \quad \text{Discharge coefficient}$$

$$A_{vent} := 0,27 \text{ m}^2 \quad \text{Area of vent}$$

$$g := 9,81 \frac{\text{m}}{\text{s}^2}$$

$$H := 1,2 \text{ m} \quad \text{Height difference between intake and vent}$$

$$T := 293 \text{ K} \quad \text{Average absolute Temperature assumed to be } 20^\circ\text{C}$$

$$V_{vent} := C \cdot A_{vent} \cdot \sqrt{2 \cdot g \cdot H \cdot \frac{\Delta T}{T}} \quad \text{Bouyancy driven ventilation estimation using stack effect formula} \quad (3.1)$$

$$V_{vent} = 0,1027 \frac{\text{m}^3}{\text{s}}$$

$$V_{fans} := 0,05 \frac{\text{m}^3}{\text{s}} \quad \text{Volumetric flowrate of two intake fans}$$

$$c_p := 1005 \frac{\text{J}}{\text{kg K}} \quad \text{Heat capacity of air}$$

$$\alpha := \frac{c_p \cdot \rho}{3600} = 0,335 \frac{\text{Pa}}{\text{K}} \quad \text{This approximation is commonly used in HVAC calculations to estimate the sensible heat removed through ventilation.} \quad (3.2)$$

$$Q_{cool} := \alpha \cdot (V_{vent} + V_{fans}) \cdot \Delta T \quad \text{Sensible cooling load due to airflow} \quad (3.3)$$

$$Q_{cool} = 0,2557 \text{ W} \quad \text{Removing } \approx 0.25 \text{ joules of heat per second}$$

4. Lighting:

$$PPFD_{required} := 400 \text{ micron} \frac{\text{mol}}{\text{m}^2 \text{ s}} \quad \text{to supply a range of needs of vegetative growth from seedlings to leafy greens}$$

$$PPFD_{sunlight} := 200 \text{ micron} \frac{\text{mol}}{\text{m}^2 \text{ s}} \quad \text{light intensity of sunlight averaged over 12 hour period assuming sunlight averages } 200 \mu\text{mol/m}^2/\text{s}$$

$$PPFD_{LED} := PPFD_{required} - PPFD_{sunlight} = 200 \text{ micron} \frac{\text{mol}}{\text{m}^2 \text{ s}} \quad \text{PPFD needed to match required level}$$

$$Grow_Area := 850 \text{ mm} \cdot 850 \text{ mm} \quad \text{estimated maximum plant coverage within greenhouse}$$

$$Light_Efficiency := 1,8 \text{ micron} \frac{\text{mol}}{\text{J}} \quad \text{micromols of PAR produced per Joule of electricity consumed assuming efficiency of LED to be } 1,8 \mu\text{mol/J}$$

$$Power_{LED} := PPFD_{LED} \cdot \frac{Grow_Area}{Light_Efficiency} = 80,2778 \text{ W} \quad \text{LED power required to supplement lighting} \quad (4)$$

5. Perspex Wind Loading:

$$t := 3 \text{ mm} \quad \text{Plate thickness} \quad \sigma_m := 72 \text{ MPa} \quad \text{Tensile strength of Plexiglass from datasheet sourced through Maizey Plastics}$$

$$v := 0,35 \quad \text{Poissons ratio} \quad b = 0,85 \text{ m}$$

$$E_{perspex} := 3,2 \text{ GPa}$$

$$v_{wind} := 40 \frac{\text{km}}{\text{hr}} \quad \text{Maximum gust winds asumed to be up to 40km/h and acting normal to panel}$$

$$D := \frac{t^3 \cdot E_{perspex}}{12 \cdot (1 - v^2)} = 8,2051 \text{ N m} \quad \text{Flexural Rigidity of the Perspex plate asuming that it is uniformly loaded and simply supported at all ends} \quad (5.1)$$

$$P := 0,5 \cdot \rho \cdot v_{wind}^2 = 74,0741 \text{ Pa} \quad \text{Dynamic pressure} \quad (5.2)$$

$$y_{pmax} := P \cdot \frac{b^4}{D \cdot 64} = 0,0736 \text{ m} \quad \text{Maximum deflection in a uniformly loaded simply supported rectangular plate using Kirchhoff-Love and Mindlin-Reissner Plate theory} \quad (5.3)$$

$$\sigma_{pmax} := \frac{6 \cdot P \cdot b^2}{t^2} = 35,679 \text{ MPa} \quad \text{Maximum bending stress in a uniformly loaded simply supported rectangular plate using Kirchhoff-Love and Mindlin-Reissner Plate theory} \quad (5.4)$$

Tensile strength of perspex $\approx 70 \text{ MPa}$

$$S_{perspex} := \frac{\sigma_m}{\sigma_{pmax}} = 2,018 \quad \text{Factor of safety for perspex due to optimistic wind loading at 40km/h} \quad (5.5)$$

6. Aluminium Frame Structure:

$$I := 4,92 \cdot 10^{-9} \text{ m}^4$$

Moment of inertia

$$c := 10 \text{ mm}$$

Distance from centroid to end of aluminium cross section

$$E := 69 \text{ GPa}$$

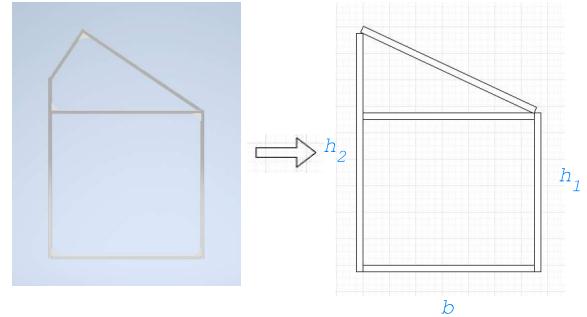
Modulus of elasticity from Macsteel

$$\sigma_y := 310 \text{ MPa}$$

Aluminium yield strength

$$C_d := 1,2$$

Drag coefficient



$$A_{side} := b \cdot h_1 + b \cdot \frac{h_2 - h_1}{2}$$

Area of side panel

$$A_{side} = 0,7735 \text{ m}^2$$

$$A_{section} := 80 \text{ mm}^2$$

Area of aluminium square tubing section

$$v_{wind} := 40 \frac{\text{km}}{\text{hr}}$$

Maximum gust winds assumed to be up to 40km/h

$$\rho_{aluminium} := 2,7 \frac{\text{g}}{\text{m}^3}$$

Density of aluminium from Macsteel

$$P := 0,5 \cdot \rho \cdot v_{wind}^2 \cdot C_d = 88,8889 \text{ Pa}$$

Pressure on panels due to wind loading (5.2)

$$F := P \cdot A_{side} = 68,7556 \text{ N}$$

Total force on side panel

$$w_{wind} := \frac{F}{2 \cdot h_1} = 41,9241 \frac{\text{N}}{\text{m}}$$

Force per meter of height due to wind

For aluminium section A-B assume that this member resists half of the force distributed across the entire side panel as a uniform load

Considering the vertical bar to be simply supported and resist uniform load. The max moment occurs at the midpoint of the bar

$$M_{wind} := w_{wind} \cdot \frac{h_1^2}{8} = 3,5237 \text{ N m}$$

Simple supports and uniform load from Shigleys Mechanical Design 10th edition

(6.1)

$$\sigma_{wind} := M_{wind} \cdot \frac{c}{I} = 7,162 \text{ MPa}$$

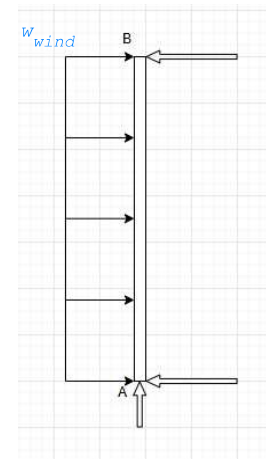
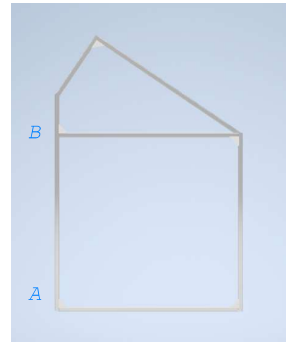
Max stress due to uniform wind load

(6.2)

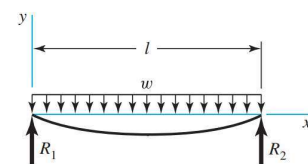
$$y_{max} := \frac{-5 \cdot w_{wind} \cdot h_1^4}{384 \cdot E \cdot I} = -0,0007 \text{ m}$$

Maximum deflection at midpoint of bar

(6.3)



7 Simple supports—uniform load



$$m_{frame} := 0,3 \frac{\text{kg}}{\text{m}} \cdot \left(2 \cdot \left(h_1 + h_2 + b + \sqrt{b^2 + (h_1 - h_2)^2} \right) + 5 \cdot l \right) = 3,6233 \text{ kg} \quad \text{Mass of all aluminium sections}$$

$$A := l \cdot h_1 \cdot 2 + l \cdot h_2 + \left(2 \cdot b \cdot \frac{(h_2 - h_1)}{2} + 2 \cdot b \cdot h_1 \right) \quad \text{Area of all sides of perspex}$$

$$V := A \cdot t = 0,0126 \text{ m}^3 \quad \text{Volume of all perspex}$$

$$m_{panels} := V \cdot \rho_{perspex} = 14,9476 \text{ kg}$$

$$m_{total} := m_{frame} + m_{panels} = 18,5709 \text{ kg} \quad \text{Total mass of structure} \quad (6.3)$$

$$F_{weight} := \frac{m_{total}}{4} \cdot 9,81 \frac{\text{m}}{\text{s}^2} = 45,5451 \text{ N} \quad \text{Assuming each vertical aluminium member supports a fourth of the total weight.} \quad (6.4)$$

$$\sigma_{weight} := \frac{F_{weight}}{A_{section}} = 0,5693 \text{ MPa} \quad \text{Stress due to weight of structure} \quad (6.5)$$

$$\sigma_{total} := \sigma_{wind} + \sigma_{weight} = 7,7314 \text{ MPa} \quad \text{Yield strength of aluminium: 100–250 MPa} \rightarrow \text{well within limits}$$

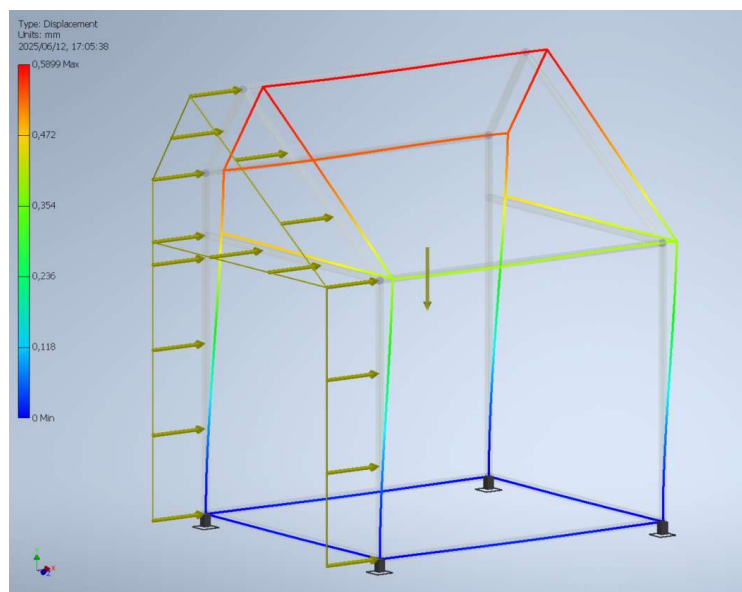
$$S_{aluminium} := \frac{\sigma_y}{\sigma_{total}} = 40,0965 \quad \text{Factor of safety for horizontal aluminium sections under wind loading}$$

6.1 Frame Analysis Using Inventor Enviroment:

$$Load_{wind} := 0,0038 \text{ N m}$$

For continuous load of 18N over all 4739mm of tubing on the side profile due to a 20km/h wind acting perpendicular to the panel. assuming 4 fixed points at the base of the greenhouse. assuming the force is resisted equally by each member on the side of the greenhouse for use in frame analysis simulation.

Maximum displacment is located at the apex of the greenhouse is limited to only 0.51mm



7.1 Fan Sizing:

$$V := b \cdot h_2 \cdot l - \frac{(h_2 - h_1)}{2} \cdot b \cdot l = 0,7735 \text{ m}^3 \quad \text{Total volume of greenhouse}$$

$$flow := \frac{V}{min} = 0,7735 \frac{\text{m}^3}{min} \quad \text{Total flow required by fans using standard airflow sizing of one entire volume exchange per minute} \quad (7.1)$$

$$CFM := flow \cdot 35 \frac{ft^3}{m^3} = 27,0725 \frac{ft^3}{min} \quad \text{Total flow in CFM}$$

7.2 HAF Requirements:

$$A_{floor} := b \cdot l = 0,85 \text{ m}^2 \quad \text{Total usable area of greenhouse}$$

$$A_{floorft} := A_{floor} \cdot 10,764 \frac{ft^2}{m^2} = 9,1494 \text{ ft}^2$$

$$CFM_{HAF} := A_{floorft} \cdot 2 \frac{ft^3}{min} = 18,2988 \frac{ft^3}{min} \quad \text{CFM Required using standard HAF sizing of 2CFM fan capacity per square foot of greenhouse area} \quad (7.2)$$

8. Power Supply:

8.1 Battery sizing:

Components	Power	Usage	Wh	
		24 hr	24 hr	
Heater	120 W	2 hr	240 Whr	~2 hours (nighttime)
Grow – Light	100 W	3 hr	300 Whr	~3 hours (morning/afternoon)
Ventilation – Fans	10 W	8 hr	80 Whr	~8 hours (daytime)
Sensors – Actuators – Controller	10 W	24 hr	240 Whr	~24 hours (intermittent use)
TOTAL	■	■	860 Whr	

$$Draw := 860 \text{ W hr}$$

$$Voltage := 12 \text{ V}$$

$$DoD := 80 \%$$

Allowable Depth of Discharge

$$Capacity := \frac{Draw}{Voltage \cdot DoD} = 89,5833 \text{ A hr} \quad \text{Minimum required battery capacity} \quad (8.1)$$

8.2 Solar Panel Sizing:

$$Battery_{Ah} := 100 \text{ A hr} \quad \text{Selecting a 100Ah battery}$$

$$hours_{day} := 6 \text{ hr} \quad \text{Assuming 6 hours of sunlight per day}$$

$$W_{panel} := Battery_{Ah} \cdot \frac{Voltage}{hours_{day} \cdot 0,8} = 250 \text{ W} \quad \text{Assuming efficiency of 80\%} \quad (8.2)$$

A solar array rated to provide atleast 250W would be able to charge the battery fully in one day assuming a moderate system efficiency of 80%

E Greenhouse Setup Images



(a) MPPT Controller



(b) LED Lighting



(c) Soil Sensor



(d) PV Panels



(e) Side profile with active vent



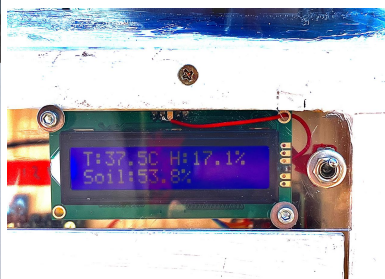
(f) Setup Overview



(g) Misting System



(h) Control Room



(i) Local Monitoring

Figure E.1: Overview of greenhouse hardware components and setup.

F Responsible use of resources and an end-of-life strategy

The responsible use of resources and a well-defined end-of-life strategy are crucial to the project. Without a sustainable plan for decommissioning, the benefits of the project, such as advancing research in renewable energy and promoting food sustainability through small-scale greenhouses, could be overshadowed by the environmental impact of improperly discarded materials. During the design and construction phase of the project, opportunities to reuse previously used materials or use recycled material were taken into consideration.

F.1 Resource Consumption Overview

The greenhouse consisted of several integrated subsystems. The physical structure, made from an aluminium frame and Perspex panels, provided an enclosed environment necessary to establish a stable greenhouse microclimate.

The system incorporated multiple electronic components, including sensors, a microcontroller, actuators, and supporting wiring and interface elements, to enable precise control of internal conditions. Water was used for irrigation, providing accurate regulation of soil moisture, humidity, and temperature through controlled misting.

Electrical power was supplied by a solar panel, battery, and charge controller, ensuring fully off-grid operation powered entirely by renewable energy sources.

F.2 Strategies for Resource Conservation and Environmental Impact Reduction

The greenhouse was designed to operate entirely on solar energy, relying solely on electricity generated from renewable resources. Component selection prioritized energy efficiency to ensure low resource consumption and long-term compatibility with off-grid solar operation.

Sensor feedback allowed actuators to operate only when necessary, conserving power and extending actuator lifespan. Although mains water was used to supply the misting system with sufficient pressure, smart control logic implemented a duty-cycle misting pattern. This ensured short, precise watering phases, preventing overwatering caused by delayed sensor feedback or control timing. The misting nozzles were positioned to target specific areas of the growth medium, further improving water efficiency.

These strategies ensured that only the required amount of water was used, a crucial consideration in a water-scarce climate like South Africa. The ventilation system also emphasized passive ventilation, using the natural buoyancy effect to exhaust warm air, see appendix D. fans were activated only when necessary to assist in airflow requirements in conjunction with the passive ventilation techniques.

Finally, durable and recyclable materials were selected to extend component life, reduce maintenance and replacement requirements, and support material reuse in future projects. These design choices align with principles of responsible resource use and sustainable end-of-life management.

F.3 End-of-Life Strategies

Once completed, the greenhouse can serve as a model to demonstrate the potential of engineering projects, inspire the adoption of renewable energy solutions, and promote education on food sustainability and self-sufficiency. Once this post completion life cycle is over. The Project can be broken down and stripped of its parts. Some of the parts will go into storage to be reused for future engineering Thesis topics or other experiments to be used by the university at their own discretion. Other parts that are less likely to be reused are to be thrown out responsibly or recycled where possible. Special attention should be paid to proper disposal of hazardous materials.

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